Multirate Distributed Receding Horizon Reinforcement Learning for Optimal UAV–UGV Formation Control

Xinglong Zhang, *Member, IEEE*, Cong Li, Ronghua Zhang, Quan Xiong, Wei Jiang, and Xin Xu, *Senior Member, IEEE*

***Abstract*—The coordination of unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs) is valuable in many applications, such as emergency search and rescue, and has received increasing attention in recent years. Given their distinct tasks and dynamic characteristics, the UAV team is typically controlled with higher maneuverability for rapid searching, while the UGV team is operated on roads at lower speeds to ensure stability and performance. This discrepancy naturally results in a multirate control problem, which has not been adequately addressed in previous works. Therefore, we present a multirate distributed receding horizon reinforcement learning (RHRL) framework to solve the optimal UAV–UGV formation control problem on fast and slow time scales. The proposed approach includes a distributed RHRL algorithm operating at a slower time scale for the formation control of UGV teams, and another distributed RHRL algorithm functioning at a faster time scale for the formation control of UAV teams. The state information among homogeneous UAV/UGV agents and heterogeneous agents across different teams are exchanged at different frequencies to balance control performance and communication load. Notably, our approach integrates the receding horizon strategy to enhance learning efficiency and provides theoretical guarantees in multirate distributed RL. Theoretically, learning convergence at different time scales and closed-loop stability are guaranteed. Comparative numerical validations are conducted to demonstrate the effectiveness of our approach in heterogeneous UAV–UGV formation control under different time scales and tasks.**

***Impact Statement*—The coordination of UAVs and UGVs is crucial in applications such as emergency search and rescue. Recent studies have addressed the UAV–UGV formation control problem using a single time scale, which would potentially impede performance due to the differing dynamic characteristics of UAVs and UGVs. The proposed approach overcomes the above challenge by introducing two distributed receding horizon reinforcement learning algorithms operating at fast and slow time scales. This method effectively resolves the time scale selection issue within the framework of multiagent and distributed reinforcement learning, achieving balanced performance for UAVs and UGVs while reducing communication load. This advancement suggests a multirate coordination mechanism in policy learning for the cooperative control of heterogeneous dynamical systems and provides a foundation for designing multirate distributed RL algorithms with theoretical guarantees.**

***Index Terms*—Multirate control, receding horizon control, reinforcement learning, UAV–UGV formation.**

I. Introduction

THE coordination of unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs) typically involves maneuvering both UAVs and UGVs through communication to accomplish cooperative tasks, showing promise in applications such as reconnaissance, surveillance, and rescue operations, particularly in large areas [1], [2]. In UAV–UGV coordination systems, UAVs typically operate at high speeds to achieve real-time perception of dynamic environments over large areas, collaboratively accomplish complex cooperative tasks, and respond promptly to emergency events. Conversely, UGVs often serve as mission execution platforms, following smoother paths at slower speeds to maintain functionality and stability. Additionally, the open-loop dynamic characteristics of UAVs differ significantly from those of UGVs [3]. For instance, UAV controllers need to react at higher frequencies in the horizontal plane to compensate for pitch angle variations, whereas UGVs can be controlled with comparatively lower maneuverability at a slower time scale, as the vertical direction effects on UGVs are often minor in most road conditions. The distinct tasks and dynamic characteristics of UAV and UGV systems naturally result in a multirate control problem, where UAVs operate at a fast time scale for highly dynamic tasks, and UGVs operate at a slow time scale to achieve smooth control objectives. However, most related works investigate heterogeneous UAV–UGV systems under a single time scale setting [4], [5]. Ignoring multirate effects can lead to performance degradation and even instability issues. While multirate control has been widely addressed in process control research to balance dynamic and long-term goals [6], no insight has been provided for the multirate control of air-ground systems.

Received 26 July 2024; revised 26 June 2025; accepted 31 August 2025. Date of publication 9 September 2025; date of current version 27 February 2026. This work was supported in part by the National Natural Science Foundation of China under Grant T2521006, Grant 62533021, Grant 62403480, Grant U24A20279, and Grant 62306325, and in part by the Science and Technology Innovation Program of Hunan Province under Grant 2024RC3145. This article was recommended for publication by Associate Editor Yumi Iwashita upon evaluation of the reviewers' comments. *(Corresponding author: Cong Li.)*

The authors are with the College of Intelligence Science and Technology, National University of Defense Technology, Changsha 410073, China (e-mail: zhangxinglong18@nudt.edu.cn; lc@nudt.edu.cn; zhangronghua19@nudt.edu.cn; xiongquan21@nudt.edu.cn; weijiang@nudt.edu.cn; xinxu@nudt.edu.cn).

Digital Object Identifier 10.1109/TAI.2025.3607722

2691-4581 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

Thereby, we address for the first time the multirate optimal control problem for heterogeneous UAV–UGV systems, an open problem that requires comprehensive exploration toward practical applications. In particular, we present a novel multirate distributed receding horizon reinforcement learning (RHRL) approach to efficiently learn the optimal control policies for UAV and UGV teams operating at different time scales.

A. Related Works

The coordination control of UAVs and UGVs is nontrivial given that the dynamic motions of aerial and ground platforms are described by different characteristics. Among the related works in the UAV–UGV coordination, the heterogeneous air-ground system consisting of a single UAV and a single UGV is investigated in [7], [8]. Furthermore, the formation control problem including multiple UAVs and UGVs is addressed under the linear simplified dynamics [9], [10] or the nonlinear dynamics [11]. In addition, centralized [12], decentralized [13], and distributed control strategies [1], [14] are proposed, respectively, for the formation of heterogeneous UAV–UGV systems. Moreover, the fault-tolerant formation control problem for heterogeneous air-ground systems is explored in [15], [16] in the presence of multiple actuator faults. As mentioned above, although much effort has been devoted to the UAV–UGV formation control from different perspectives, the multirate control issue is rarely discussed.

The multirate control is prevalent in modern industrial processes, consisting of multiple unit processes that operate on a fast time scale and the operation process operating on a slow time scale [17], [18]. To enhance dynamic performance, the multirate control of the industrial process is studied via learning or control perspectives [19], [20], [21], [22], [23]. The extension of the multirate control to the robotics is explored in [24], wherein a multirate tracking control strategy for a space robot on a controlled satellite is proposed to address the stability issue arising from different sampling rates of the manipulator and the base controllers. Recent works [25], [26], [27] creatively use the concept of multirate to describe the interplay between planning and control layers, which work at different frequencies. The promising work [28] develops a data-driven control approach for the asynchronous control problem of the multirate sampled-data nonlinear systems. However, to the best of our knowledge, no prior work has extended the multirate control scheme to the UAV–UGV formation control problem.

The optimal control of UAVs or UGVs [29], [30] has been thoroughly investigated separately; however, the optimal control of the UAV–UGV formation remains to be fully explored, particularly in the context of the multirate control problem. Model predictive control (MPC) serves as the representative optimal control method to solve the above problem by casting the control problem as an optimization problem to be solved online [31], [32]. However, the online computation of distributed nonlinear MPC is intractable [33], especially for medium or large-scale systems.

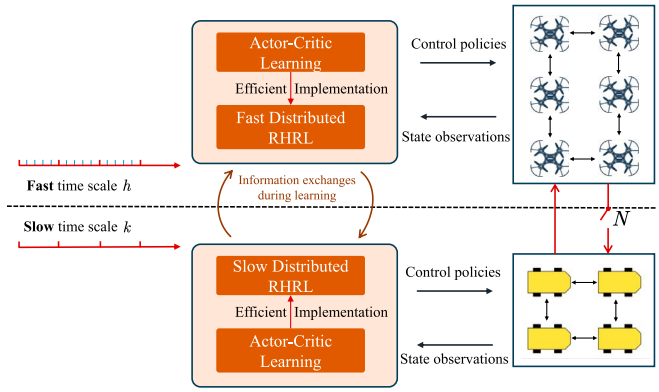


Fig. 1. Schematic diagram of the proposed multirate distributed RHRL approach. A slow distributed RHRL approach works on the slow time scale k for the UGV formation, while a fast distributed RHRL algorithm works on the fast time scale h for the UAV formation. The distributed RHRL approaches are implemented by the corresponding distributed actor-critic algorithms at fast and slow time scales. The information exchanges at each time instant and at each prediction horizon are allowed for coordinated policy optimization between UAV and UGV teams.

Distributed and multiagent reinforcement learning is promising in designing cooperative control policies for heterogeneous MASs [34], [35], [36], which, however, suffers from the nonstationary issue, low learning efficiency, and lacks theoretical guarantees. Receding horizon RL (RHRL), as a novel approach integrating the MPC theory into RL, serves as a promising method to learn closed-loop control policies with guarantees in the centralized setting [37], [38] and the distributed setting [39]. In this article, we develop multirate distributed RHRL to solve the optimal UAV–UGV formation control problem on different time scales. To the best of our knowledge, no previous work has addressed the multirate control problem of the heterogeneous MASs via the distributed RL approach.

B. Contribution

This article proposes a multirate distributed RHRL approach for optimal UAV–UGV formation control at fast and slow time scales. The proposed approach consists of a distributed RHRL algorithm that works at a slow time scale for the formation control of UGV teams and a distributed RHRL algorithm that works at a fast time scale for the formation control of UAV teams. The proposed approach is efficiently implemented online by a novel multirate distributed actor-critic learning algorithm performed in an incremental learning fashion (see Fig. 1 for a schematic diagram of our control scheme). Our main contributions are summarized as follows.

1) We investigate for the first time a multirate cooperative control problem for the UAV–UGV formation, in which the UAVs and UGVs are working at fast and slow time scales, respectively, with different formation control goals.
2) We propose a multirate distributed RHRL framework to generate the optimal formation control policies for UAVs and UGVs at different time scales. The receding horizon strategy rooted in MPC is integrated into policy learning,

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

which enhances learning efficiency and provides theoretical guarantees. Unlike multirate MPC methods in the process control community, our approach learns explicit closed-loop control policies rather than open-loop control sequences. Consequently, the learned policies for UAVs and UGVs can be transferred online to different control scenarios.

3) We present a multirate distributed actor-critic algorithm to efficiently implement the proposed approach. To the best of our knowledge, no previous actor-critic RL approach with a multirate learning mechanism has been developed.

The remainder of this article is structured as follows. The multirate formation control problem is presented in Section II, while the proposed multirate distributed RHRL framework, as well as the theoretical guarantees, are provided in Section III. The multirate distributed actor-critic implementation is given in Section IV. Numerical simulation results on UAV–UGV formation control are shown in Section V, demonstrating the effectiveness of our approach. Finally, conclusions are drawn in Section VI. Proofs of the main results and the parameter setting in the simulation are given in Appendix.

Notation: Throughout this article, we use $\mathbb{R}$ and $\mathbb{R}^+$ to denote sets of real numbers and positive real numbers, respectively; $\mathbb{R}^n$ to denote the Euclidean space of the n-dimensional real vector; $\mathbb{R}^{n\times m}$ to denote the Euclidean space of $n\times m$ real matrices. Denoting $\mathbb{N}$ as the set of integers; $\mathbb{N}_{l_1}^{l_2}$ as the set of integers $l_1, l_1+1,\ldots,l_2$. We use the notation $\lfloor k/N\rfloor$ to denote the largest integer less than k/N. For a group of vectors $z_i\in\mathbb{R}^{n_i}$, $i\in\mathbb{N}_1^M$, we use $\text{col}_{i\in\mathbb{N}_1^M}(z_i)$ or $(z_1,\ldots,z_M)$ to denote $[z_1^\top,\ldots,z_M^\top]^\top$, where $M\in\mathbb{N}$. For a general function $h(z(k))$ in a variable $z(k)$, we use $h(k)$ to represent $h(z(k))$ for simplicity. For a matrix $P\in\mathbb{R}^{n\times n}$, $P\succ 0$ means that it is positive definite. For a vector $x\in\mathbb{R}^n$, we denote $\|x\|_Q^2$ as $x^\top Qx$ and $\|x\|$ as the Euclidean norm. For notational brevity, time-dependence is suppressed without causing ambiguity. The Kronecker product of two general matrices A and B is denoted as $A\otimes B$.

II. Problem Formulation

In this section, we first present the dynamic models of the UAV and UGV systems and derive the associated formation error models at fast and slow time scales, respectively. Next, we formulate the multirate optimal UAV–UGV formation control problem, in which two distributed MPC problems are stated for UAVs and UGVs at different time scales.

A. UAV/UGV Models at Fast and Slow Time Scales

Consider the formation control of a group of M_a UAVs and a group of M_g UGVs. The dynamical model of each UAV $i\in\mathbb{N}_1^{M_a}$ in continuous time is described as

$$\begin{cases}\dot{x}_{a,i}=v_{a,i}\cos\gamma_{a,i}\cos\varphi_{a,i}\\ \dot{y}_{a,i}=v_{a,i}\cos\gamma_{a,i}\sin\varphi_{a,i}\\ \dot{z}_{a,i}=v_{a,i}\sin\gamma_{a,i}\\ \dot{\gamma}_{a,i}=\mu_{a,i}\\ \dot{\varphi}_{a,i}=\omega_{a,i}\\ \dot{v}_{a,i}=a_{a,i}\end{cases}\tag{1}$$

where $x_{a,i}, y_{a,i}, z_{a,i}\in\mathbb{R}$ are the positions in the Cartesian coordinate frame; $\gamma_{a,i}, \varphi_{a,i}\in\mathbb{R}$ are the pitch and yaw angles; $v_{a,i}, a_{a,i}\in\mathbb{R}$ are the velocity and accelerate; $\mu_{a,i}, \omega_{a,i}\in\mathbb{R}$ are the pitch rate and yaw rate, respectively.

The dynamical model of each UGV $i\in\mathbb{N}_1^{M_g}$ in continuous time is described as

$$\begin{cases}\dot{x}_{g,i}=v_{g,i}\cos\varphi_{g,i}\\ \dot{y}_{g,i}=v_{g,i}\sin\varphi_{g,i}\\ \dot{\varphi}_{g,i}=\omega_{g,i}\\ \dot{v}_{g,i}=a_{g,i}\end{cases}\tag{2}$$

where $x_{g,i}, y_{g,i}\in\mathbb{R}$ are positions of UGV in the horizontal plane; $\varphi_{g,i}, \omega_{g,i}\in\mathbb{R}$ are the yaw angle and yaw rate; $v_{g,i}, a_{g,i}\in\mathbb{R}$ are the velocity and acceleration, respectively.

This article considers a nontrivial formation control problem with heterogeneous UAV and UGV teams, wherein the UAV team and the UGV team have their own formation goals. In practical applications, the UAV team is usually tasked with searching or patrolling at higher speeds, while the UGV team runs at a lower speed to follow some smooth paths. In this setting, UAV controllers must react immediately to task or environmental changes, while the UGV can be regulated smoothly. This naturally leads to a multirate control problem in which the UAV team is controlled at a basic time scale h with a sampling interval Δt, while the UGV team is controlled on a slower time scale k with an integer N such that $h=kN$.

To this end, we first discretize the dynamical models of UAV and UGV at the basic time scale h. Then, the UGV models are resampled with a period N to derive the models corresponding to the slow time scale k. The models of UAV and UGV at the time scale h are given as follows:

$$s_{a,i}(h+1)=\bar{f}_{a,i}(s_{a,i}(h))+\bar{g}_{a,i}(s_{a,i}(h))\bar{u}_{a,i}(h),\ \forall i\in\mathbb{N}_1^{M_a}\tag{3a}$$

$$s_{g,i}(h+1)=\bar{f}_{g,i}(s_{g,i}(h))+\bar{g}_{g,i}(s_{g,i}(h))\bar{u}_{g,i}(h),\ \forall i\in\mathbb{N}_1^{M_g}\tag{3b}$$

where $s_{a,i}=(x_{a,i},y_{a,i},z_{a,i},\gamma_{a,i},\varphi_{a,i},v_{a,i})\in\mathbb{R}^6$ are the states of ith UAV, $\bar{u}_{a,i}=(\mu_{a,i},\omega_{a,i},a_{a,i})\in\mathbb{R}^3$ are the control inputs, $\bar{f}_{a,i}:\mathbb{R}^6\to\mathbb{R}^6$ and $\bar{g}_{a,i}:\mathbb{R}^6\to\mathbb{R}^{6\times 3}$ are the associated state transition and input mapping functions in discrete time, respectively; $s_{g,i}=(x_{g,i},y_{g,i},\varphi_{g,i},v_{g,i})\in\mathbb{R}^4$ are the states of ith UGV, $\bar{u}_{g,i}=(\omega_{g,i},a_{g,i})\in\mathbb{R}^2$ are the control inputs, $\bar{f}_{g,i}:\mathbb{R}^4\to\mathbb{R}^4$ and $\bar{g}_{a,i}:\mathbb{R}^4\to\mathbb{R}^{4\times 2}$ are associated state transition and input mapping functions in discrete time, respectively.

Let $\mathcal{N}_{\star,i}$ be the set of the ith agents' neighbors within their team (see Fig. 2) and denote $s_{\mathcal{N}_{\star,i}}$ as the corresponding neighboring state of agent i, for $\star=a,g$. Let $\mathcal{N}_{a\to g,i}$ be the set of ith UGVs' neighbors in the UAV team, and denote $s_{\mathcal{N}_{a\to g,i}}$ as the corresponding neighboring state of agent i. The notation $\mathcal{N}_{g\to a,i}$ and $s_{\mathcal{N}_{g\to a,i}}$ are defined analogously. Before introducing the formation control goals of UAVs and UGVs, we provide some basic assumptions on the communication protocol as follows.

Assumption 1 (Communication protocol):

1) In the UAV team, the communications among the neighboring UAVs are allowed at each basic time scale h, i.e., for any $i\in\mathbb{N}_1^{M_a}$, the state information $s_{\mathcal{N}_{a,i}}$ can be transmitted to the ith UAV at each time h.

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

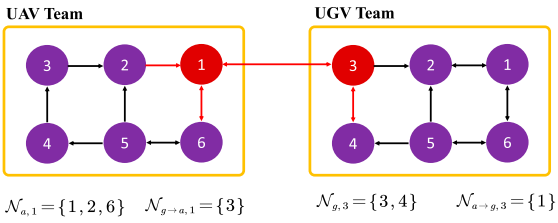


Fig. 2. Illustrative example of the definition of neighbors in formation control of 6 UAVs and 6 UGVs. The directed lines with arrows represent the directions of the information flows. For the first UAV, the set of neighbors from its own team is $\mathcal{N}_{a,1}=\{1,2,6\}$, the set of neighbors from the UGV team is $\mathcal{N}_{g\to a,1}=\{3\}$. For the 3rd UGV, the set of neighbors from its own team is $\mathcal{N}_{g,3}=\{3,4\}$, the set of neighbors from the UAV team is $\mathcal{N}_{a\to g,3}=\{1\}$.

2) In the UGV team, the communications among the neighboring UGVs are allowed at each slow time scale k, i.e., for any $i\in\mathbb{N}_1^{M_g}$, the state information $s_{\mathcal{N}_{g,i}}$ can be transmitted to the ith UGV at each time instant $h=kN$.
3) Communications between the UAV and UGV teams are allowed. The state information $s_{\mathcal{N}_{g\to a,i}}$ from the UGV team to the UAV team is transmitted at each time instant h, while the state information $s_{\mathcal{N}_{a\to g,i}}$ from the UAV team to the UGV team is transmitted at each time instant k (corresponding to $h=kN$ at the basic time scale).

The above assumption about communications is standard since only communications among neighbors are allowed among agents. Moreover, the information to the UGVs is only transmitted within the slow time scale as needed, reducing the communication load.

In order to design the UGV controller at the slow time scale $k\in\mathbb{N}$ with $k=\lfloor h/N\rfloor$, we denote as $u_{g,i}^{[N]}$ and $s_{g,i}^{[N]}$ the samplings of $u_{g,i}$ and $s_{g,i}$ corresponding to the time scale k. Hence, the sampled model of (3b) with N period follows:

$$\begin{aligned}s_{g,i}^{[N]}(k+1)=\ &\bar{f}_{g,i}(s_{g,i}^{[N]}(k))+\sum_{l=1}^{N-1}\bar{f}_{g,i}(s_{g,i}(h+l))\\&+\bar{g}_{g,i}(s_{g,i}(h+l))u_{g,i}^{[N]}(k).\end{aligned}\tag{4}$$

Now, we introduce the definition of formation errors for the coordination of UAVs and UGVs. The formation error of the UAV in the local coordinate frame is defined at the basic time scale h according to the communication protocol, i.e.,

$$\begin{aligned}e_{a,i}=\Theta_{a,i}\Bigg(&\sum_{j\in\mathcal{N}_{a,i}}(\Lambda_{a,1}(s_{a,j}-s_{a,i})+\rho_{a,ji})\\&+c_{a,i}(\Lambda_{a,1}(s_{a,r}-s_{a,i})+\rho_{a,ri}+\Lambda_{a,2}(s_{a,r}-s_{a,i}))\\&+\sum_{j\in\mathcal{N}_{g\to a,i}}\Lambda_{a,1}(C_g s_{g,j}-s_{a,i})\Bigg)\end{aligned}\tag{5}$$

where $c_{a,i}=1$ if ith UAV receives the position information of the leader and $c_{a,i}=0$ otherwise, $\Lambda_{a,1}=\mathrm{diag}\{1,1,0,0,0,0\}$, $\Lambda_{a,2}=\mathrm{diag}\{0,0,1,1,1,1\}$, $s_{a,r}$ is the reference state from the leader; $\rho_{a,ji}$ and $\rho_{a,ri}$ are coordinate correction variables, which are determined by the formation shape and size; $C_g\in\mathbb{R}^{6\times4}$ projects the UGV state space $\mathbb{R}^4$ to the UAV state space $\mathbb{R}^6$; the coordinate transformation matrix is

$$\Theta_{a,i}=\begin{bmatrix}\cos\varphi_{a,i}&\sin\varphi_{a,i}&0\\-\sin\varphi_{a,i}&\cos\varphi_{a,i}&0\\0&0&I_4\end{bmatrix}.$$

Let $u_{a,i}=\bar{u}_{a,r}-\bar{u}_{a,i}\in\mathbb{R}^3$ be the control variable associated with the formation goal of the ith UAV, where $\bar{u}_{a,r}$ consist of the reference controls from the leader. Denote $e_{\mathcal{N}_{a,i}}\in\mathbb{R}^{n_{\mathcal{N}_{a,i}}}$ as the collection of all neighboring error states (including $e_{a,i}$) in the UAV team, i.e., $e_{\mathcal{N}_{a,i}}=\mathrm{col}_{j\in\mathcal{N}_{a,i}}e_{a,j}$, and denote $e_{\mathcal{N}_{g\to a,i}}\in\mathbb{R}^{n_{\mathcal{N}_{g\to a,i}}}$ as the collection of all connected error states from UGV. One can write the local formation error model of the ith UAV as follows:

$$\begin{aligned}e_{a,i}(h+1)=\ &f_{a,i}(e_{\mathcal{N}_{a,i}}(h),e_{\mathcal{N}_{g\to a,i}}(h))\\&+g_{a,i}(e_{a,i}(h))u_{a,i}(h).\end{aligned}\tag{6}$$

Analogously, we define the UGV formation error at each time instant $h=kN$ as

$$\begin{aligned}e_{g,i}=\Theta_{g,i}\Bigg(&\sum_{j\in\mathcal{N}_{g,i}}(\Lambda_{g,1}(s_{g,j}-s_{g,i})+\rho_{g,ji})\\&+c_{g,i}(\Lambda_{g,1}(s_{g,r}-s_{g,i})+\rho_{g,ri}+\Lambda_{g,2}(s_{g,r}-s_{g,i}))\\&+\sum_{j\in\mathcal{N}_{a\to g,i}}\Lambda_{g,1}(C_a s_{a,j}-s_{g,i})\Bigg)\end{aligned}\tag{7}$$

where $c_{g,i}=1$ if ith UGV receives the position information of the leader and $c_{g,i}=0$ otherwise, $\Lambda_{g,1}=\mathrm{diag}\{1,1,0,0\}$, $\Lambda_{g,2}=\mathrm{diag}\{0,0,1,1\}$; $s_{g,r}$ is the reference state received from the leader; $\rho_{g,ji}$ and $\rho_{g,ri}$ are coordinate correction variables, which are determined by the formation shape and size, $C_a\in\mathbb{R}^{4\times6}$ projects the UAV state space $\mathbb{R}^6$ to the UGV state space $\mathbb{R}^4$; the coordinate transformation matrix is

$$\Theta_{g,i}=\begin{bmatrix}\cos\varphi_{g,i}&\sin\varphi_{g,i}&0\\-\sin\varphi_{g,i}&\cos\varphi_{g,i}&0\\0&0&I_2\end{bmatrix}.$$

Let $u_{g,i}=\bar{u}_{g,r}-\bar{u}_{g,i}\in\mathbb{R}^2$ consist of the control variable associated with the formation goal of the ith UGV, where $\bar{u}_{g,r}$ are the reference control inputs received from the leader, denote $e_{\mathcal{N}_{g,i}}\in\mathbb{R}^{n_{\mathcal{N}_{g,i}}}$ as the collection of all neighboring error states (including $e_{g,i}$) in the UGV team, i.e., $e_{\mathcal{N}_{g,i}}=\mathrm{col}_{j\in\mathcal{N}_{g,i}}e_{g,j}$, and denote $e_{\mathcal{N}_{a\to g,i}}\in\mathbb{R}^{n_{\mathcal{N}_{a\to g,i}}}$ as the collection of all connected error states from UGV (not including $e_{g,i}$). One can write the local formation error model of the ith UGV at the slow time scale k as follows:

$$e_{g,i}^{[N]}(k+1)=f_{g,i}(e_{\mathcal{N}_{g,i}}^{[N]}(k),e_{\mathcal{N}_{a\to g,i}}^{[N]}(k))+g_{g,i}(e_{g,i}^{[N]}(k))u_{g,i}^{[N]}(k)\tag{8}$$

which is equivalent to the model recursion at the basic time $h=kN$, i.e.,

$$\begin{aligned}e_{g,i}(kN+N)=\ &f_{g,i}(e_{\mathcal{N}_{g,i}}(kN),e_{\mathcal{N}_{a\to g,i}}(kN))\\&+g_{g,i}(e_{g,i}(kN))u_{g,i}^{[N]}(\lfloor h/kN\rfloor).\end{aligned}$$

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

The following basic assumption on models (6) and (8) is assumed to hold.

Assumption 2 (Multirate stabilization): For the fast error model (6) and the slow error model (8), there respectively exist $u_{a,i}(e_{\mathcal{N}_{a,i}}, e_{\mathcal{N}_{g\to a,i}})$ for $i \in \mathbb{N}_1^{M_a}$ and $u_{g,i}(e^{[N]}_{\mathcal{N}_{g,i}}, e^{[N]}_{\mathcal{N}_{a\to g,i}})$ for $i \in \mathbb{N}_1^{M_g}$, such that the closed-loop system for UAV–UGV formation control is asymptotically stable.

This assumption is an extension of stabilizing control condition in [40] to the multirate control case, which is useful in deriving the closed-loop stability deferred in Section III-B.

B. Multirate UAV–UGV Formation Control Problem

In what follows, we formulate the multirate optimal UAV–UGV formation control problem with a multirate MPC framework. Two distributed MPC problems are stated for UGVs and UAVs respectively, and are solved at different time scales.

First, at each slow time instant k (corresponding to the fast time instant $h = kN$), the following finite-horizon optimization problem is to be solved for the UGV team:

$$\begin{aligned} \min_{\overrightarrow{\boldsymbol{u}}^{[N]}_g(k)} \quad & J_g(k) \\ \text{s. t.:} \quad & \text{model (8)} \end{aligned} \tag{9}$$

where $J_g(k) = \sum_{i=1}^{M_g} J_{g,i}(k)$, and

$$\begin{aligned} J_{g,i}(k) = & \sum_{l=k}^{k+N_g-1} L_{g,i}(e^{[N]}_{\mathcal{N}_{g,i}}(l), e^{[N]}_{\mathcal{N}_{a\to g,i}}(l), u^{[N]}_{g,i}(l)) \\ & + \left\| e^{[N]}_{g,i}(l+N_g) \right\|^2_{P_{g,i}} \end{aligned} \tag{10}$$

and $\boldsymbol{u}^{[N]}_g = \mathrm{col}_{i\in\mathbb{N}_1^{M_g}} u^{[N]}_{g,i}$, $\overrightarrow{\boldsymbol{u}}^{[N]}_g(k) = \boldsymbol{u}^{[N]}_g(k), \ldots, \boldsymbol{u}^{[N]}_g(k+N_g-1)$, N_g is the prediction horizon, $P_{g,i} = P^\top_{g,i} \succ 0$ is the terminal penalty matrix, whose design principles are deferred in Appendix A. The stage cost for the ith UGV is defined as

$$\begin{aligned} & L_{g,i}(e^{[N]}_{\mathcal{N}_{g,i}}(l), e^{[N]}_{\mathcal{N}_{a\to g,i}}(l), u^{[N]}_{g,i}(l)) = \left\| e^{[N]}_{\mathcal{N}_{g,i}}(l) \right\|^2_{Q_{g,i}} \\ & + \left\| e^{[N]}_{\mathcal{N}_{a\to g,i}}(l) \right\|^2_{Q_{a\to g,i}} + \left\| u^{[N]}_{g,i}(l) \right\|^2_{R_{g,i}} \end{aligned} \tag{11}$$

where $Q_{g,i} \in \mathbb{R}^{n_{\mathcal{N}_{g,i}} \times n_{\mathcal{N}_{g,i}}}$, $Q_{a\to g,i} \in \mathbb{R}^{n_{\mathcal{N}_{a\to g,i}} \times n_{\mathcal{N}_{a\to g,i}}}$, $R_{g,i} \in \mathbb{R}^{m_{g,i}\times m_{g,i}}$ are symmetric positive-definite matrices.

At each basic time scale $h = kN + q$, $q \in \mathbb{N}_0^{N-1}$, the following finite-horizon optimization problem is to be solved for the UAV team:

$$\begin{aligned} \min_{\overrightarrow{\boldsymbol{u}}_a(h)} \quad & J_a(h) \\ \text{s. t.:} \quad & \text{model (6)} \end{aligned} \tag{12}$$

where $J_a(h) = \sum_{i=1}^{M_a} J_{a,i}(h)$, and

$$\begin{aligned} J_{a,i}(h) = & \|e_{a,i}(h+N_a)\|^2_{P_{a,i}} \\ & + \sum_{\tau=h}^{h+N_a-q-1} L_{a,i}\left(e_{\mathcal{N}_{a,i}}(\tau),\ e_{\mathcal{N}_{g\to a,i}}(\tau),\ u_{a,i}(\tau)\right) \end{aligned} \tag{13}$$

$\boldsymbol{u}_a = \mathrm{col}_{i\in\mathbb{N}_1^{M_a}} u_{a,i}$, $\overrightarrow{\boldsymbol{u}}_a(h) = \boldsymbol{u}_a(h), \ldots, \boldsymbol{u}_a(h+N_a-q-1)$, $N_a - q$ is the shrinking prediction horizon satisfying $N_a = N_g N$, $P_{a,i} = P^\top_{a,i} \succ 0$ is the terminal penalty matrix which will be defined in (34) in Appendix A. The stage cost for the ith UAV is defined as

$$\begin{aligned} & L_{a,i}(e_{\mathcal{N}_{a,i}}(\tau), e_{\mathcal{N}_{g\to a,i}}(\tau), u_{a,i}(\tau)) = \left\| e_{\mathcal{N}_{a,i}}(\tau) \right\|^2_{Q_{a,i}} \\ & + \left\| e_{\mathcal{N}_{g\to a,i}}(\tau) \right\|^2_{Q_{g\to a,i}} + \left\| u_{a,i}(\tau) \right\|^2_{R_{a,i}} \end{aligned} \tag{14}$$

where $Q_{a,i} \in \mathbb{R}^{n_{\mathcal{N}_{a,i}} \times n_{\mathcal{N}_{a,i}}}$, $Q_{g\to a,i} \in \mathbb{R}^{n_{\mathcal{N}_{g\to a,i}} \times n_{\mathcal{N}_{g\to a,i}}}$, $R_{a,i} \in \mathbb{R}^{m_{a,i}\times m_{a,i}}$ are symmetric positive-definite matrices.

At each fast time scale h, the optimal control sequence is calculated by solving the optimization problem (12) for UAVs. Only the first control is applied, and (12) is solved at the next time instant $h+1$. In contrast, the optimization problem for the UGVs is solved at each slow time scale k to derive the optimal control sequence. Similarly, only the first control is applied, and (12) is solved at the next time instant $k+1$. Note that solving (9) and (12) in real time using nonlinear numerical solvers is intractable. Furthermore, solving local optimization problems in (9) and (12) requires the calculated control sequences and the predicted states of neighboring agents, which are often not identical to the actual values.

Remark 1: In this article, our proposed multirate RHRL approach decomposes the overall optimization problem over each prediction interval into several subproblems and solves them using a forward heuristic learning approach. This method allows control and state predictions among different agents to be immediately transmitted to their neighbors at each prediction time instant for efficient policy updates. Furthermore, unlike multirate MPC methods in the process control community, our approach learns explicit closed-loop control policies rather than open-loop control sequences. The learned policies for UAVs and UGVs can be transferred online to different control scenarios. The design details of our approach will be illustrated in the next section.

III. Multirate Distributed RHRL Framework

This section presents the proposed multirate distributed RHRL framework to solve the optimal UAV–UGV formation problem. In particular, a multirate distributed receding horizon value iteration algorithm is designed. Then, theoretical results on learning convergence and closed-loop stability are derived.

A. Design of Multirate Distributed RHRL

Hamilton–Jacobi–Bellman (HJB) equation at slow time scale: Assume that the optimization problem (9) has been solved in past time instants and now is to be solved at a generic slow time instant k. However, as mentioned previously, solving (9) distributedly on the prediction horizon requires the calculated control sequence and the predicted state of the neighbors in and out of the UGV team. For this reason, we decompose the optimization problem (9) into N_g sequential subproblems and solve them in a forward-in-time heuristic manner. To this end, at each time instant $l \in [k, k+N_g-1]$, denoting $J_{g,i}(l) = L_{g,i}(l) + J_{g,i}(l+1)$ and letting $J^*_{g,i}(l)$ be the optimal value function associated with the optimal control policy $u^{[N],*}_{g,i}(l)$,

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

we first write the corresponding HJB equation at the slow time scale, for $l \in [k, k+N_g-1]$, as

$$\sum_{i=1}^{M_g} J^*_{g,i}(\bar{e}^{[N]}_{\mathcal{N}_{g,i}}(l)) = \min_{u^{[N]}_{g,i}(l), i\in\mathbb{N}_1^{M_g}} \sum_{i=1}^{M_g} L_{g,i}(l) + \sum_{i=1}^{M_g} J^*_{g,i}\big(\bar{e}^{[N]}_{\mathcal{N}_{g,i}}(l+1)\big)$$

where $\bar{e}^{[N]}_{\mathcal{N}_{g,i}} = (e^{[N]}_{\mathcal{N}_{g,i}}, e^{[N]}_{\mathcal{N}_{a\to g,i}}) \in \mathbb{R}^{\bar{n}_{g,i}}$, $\bar{n}_{g,i} = n_{\mathcal{N}_{g,i}} + n_{\mathcal{N}_{a\to g,i}}$. The optimal control of the ith UGV at each time $l \in [k, k+N_g-1]$ could be calculated utilizing the related one-step ahead optimal cost $J^*_{g,j}(l+1)$, that is,

$$u^{[N],*}_{g,i}(l) = \underset{u^{[N]}_{g,i}(l)}{\operatorname{argmin}} \left\{ L_{g,i}(l) + \sum_{j\in\bar{\mathcal{N}}_{g,i}} J^*_{g,j}\big(\bar{e}^{[N]}_{\mathcal{N}_{g,j}}(l+1)\big) \right\} \tag{15}$$

for all $i \in \mathbb{N}_1^{M_g}$, $\bar{\mathcal{N}}_{g,i}$ is the set with the ith UGV as its neighbor.

HJB equation at fast time scale: Analogously, assume now that, at a generic fast time instant $h = kN+q$, $q \in \mathbb{N}_0^{N-1}$, the optimization problem (12) must be solved. At each time instant $\tau \in [h, h+N_a-q-1]$, denoting $J_{a,i}(\tau) = L_{a,i}(\tau) + J_{a,i}(\tau+1)$ and letting $J^*_{a,i}(\tau)$ be the optimal value function associated with the optimal control policy $u^*_{a,i}(\tau)$, we write the HJB equation for the UAV team, $\forall\, \tau \in [h, h+N_a-q-1]$, as

$$\sum_{i=1}^{M_a} J^*_{a,i}\big(\bar{e}_{\mathcal{N}_{a,i}}(\tau)\big) = \min_{\substack{u_{a,i}(\tau),\\ i\in\mathbb{N}_1^{M_a}}} \sum_{i=1}^{M_a} L_{a,i}(\tau) + \sum_{i=1}^{M_a} J^*_{a,i}\big(\bar{e}_{\mathcal{N}_{a,i}}(\tau+1)\big)$$

where $\bar{e}_{\mathcal{N}_{a,i}} = (e_{\mathcal{N}_{a,i}}, e_{\mathcal{N}_{g\to a,i}}) \in \mathbb{R}^{\bar{n}_{a,i}}$, $\bar{n}_{a,i} = n_{\mathcal{N}_{a,i}} + n_{\mathcal{N}_{g\to a,i}}$. The optimal control of the ith UAV at each time instant $\tau \in [h, h+N_a-q-1]$ could be calculated utilizing the related one-step ahead optimal cost $J^*_{a,j}(\tau+1)$, that is,

$$u^*_{a,i}(\tau) = \underset{u_{a,i}(\tau)}{\operatorname{argmin}} \left\{ L_{a,i}(\tau) + \sum_{j\in\bar{\mathcal{N}}_{a,i}} J^*_{a,j}(\tau+1) \right\} \tag{16}$$

for all $i \in \mathbb{N}_1^{M_a}$, $\bar{\mathcal{N}}_{a,i}$ is the set with the ith UAV as its neighbor.

Multirate receding horizon policy learning: In what follows, we propose a policy learning approach to solve (15) at the slow time scale and solve (16) at the fast time scale, which are described in Algorithm 1. The policy learning in each time interval $[kN, kN+N-1]$ is divided into two procedures. First, at the time instant $h = kN$, the problems (15) and (16) are solved in an asynchronous manner, while (16) is solely solved in the interval $h \in [kN+1, kN+N-1]$.

Algorithm 1: Multirate Distributed Policy Learning in Each Time Interval $[kN, kN+N-1]$.

Procedure 1: Policy learning at $h = kN$.

Require: $\boldsymbol{u}_g^{[N],0}(l)$, $l \in \mathbb{N}_k^{k+N_g-1}$, $\boldsymbol{u}_a^0(\tau)$, $\tau \in \mathbb{N}_{kN}^{kN+N_a-1}$, $t = 0$, $t_{\max}$;

while $t \le t_{\max}$ **do**
  for $\tau = kN, \cdots, kN+N_a-1$ **do**
    if $\operatorname{mode}(\tau, N) = 0$ **then**
      $l = \lfloor \tau/N \rfloor$;
      Value update and policy learning, for $i \in \mathbb{N}_1^{M_g}$:

$$J^{t+1}_{g,i}(l) = L_{g,i}(l) + J^t_{g,i}(l+1) \tag{17a}$$

$$u^{[N],t+1}_{g,i}(l) = \underset{u^{[N]}_{g,i}(l)}{\operatorname{argmin}} \{L_{g,i}(l) + \sum_{j\in\bar{\mathcal{N}}_{g,i}} J^{t+1}_{g,j}(l+1)\}. \tag{17b}$$

    end if
  end for
  for $\tau = kN, \cdots, kN+N_a-1$ **do**
    Value update and policy learning, for all $i \in \mathbb{N}_1^{M_a}$:

$$J^{t+1}_{a,i}(\tau) = L_{a,i}(\tau) + J^t_{a,i}(\tau+1) \tag{18a}$$

$$u^{t+1}_{a,i}(\tau) = \underset{u_{a,i}(\tau)}{\operatorname{argmin}} \{L_{a,i}(\tau) + \sum_{j\in\bar{\mathcal{N}}_{a,i}} J^{t+1}_{a,j}(\tau+1)\}. \tag{18b}$$

  end for
  $t \leftarrow t+1$;
end while

Procedure 2: Policy learning at $h \in [kN+1, kN+N-1]$.

Require: $\boldsymbol{u}_a^0(\tau)$, $\tau \in \mathbb{N}_h^{h+N_a-q-1}$, $t = 0$, $t_{\max}$;

while $t \le t_{\max}$ **do**
  for $\tau = h, \cdots, h+N_a-q-1$ **do**
    Value update and policy learning with (18);
  end for
  $t \leftarrow t+1$;
end while

B. Theoretical Analysis

In what follows, we first prove the learning convergence of Algorithm 1 with Theorems 1 and 2, i.e., the updated control policies at fast and slow time scales finally converge to the optimal values. Then, the closed-loop stability of the multirate RHRL algorithm is analyzed in Theorem 3 under the receding-horizon learning strategy.

Theorem 1 (Learning convergence at $h = kN$): Under Procedure 1 in Algorithm 1, we have

1) $J^{t+1}_a(\tau) \le J^t_a(\tau)$ for all $\tau \in [kN, kN_a-1]$; $J^{t+1}_g(l) \le J^t_g(l)$ for $l = \lfloor \tau/N \rfloor$.
2) $J^t_a(\tau) \to J^*_a(\tau)$, $\boldsymbol{u}^t_a(\tau) \to \boldsymbol{u}^*_a(\tau)$ for all $\tau \in [h, kN+N_a-1]$, as $t \to +\infty$; $J^t_g(l) \to J^*_g(l)$, $\boldsymbol{u}^{[N],t}_g(l) \to \boldsymbol{u}^{[N],*}_g(l)$ for $l = \lfloor \tau/N \rfloor$, as $t \to +\infty$.

Proof: Please see Appendix C. ■

Theorem 2 (Convergence at $h \in [kN+1, kN+N)$): Under Procedure 2 in Algorithm 1, we have

1) $J^{t+1}_a(\tau) \le J^t_a(\tau)$ for all $\tau \in [h, kN+N_a-1]$.
2) $J^t_a(\tau) \to J^*_a(\tau)$, $\boldsymbol{u}^t_a(\tau) \to \boldsymbol{u}^*_a(\tau)$ for all $\tau \in [h, kN+N_a-1]$, as $t \to +\infty$.

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

Proof: Procedure 2 only involves the policy update of the UAVs at the fast time scale. The proof arguments are analogous to those in Theorem 1. ■

In practical implementations, it is impossible to perform unlimited iteration with $t \to \infty$ to derive $J_\star^t \to J_\star^*$ with $\star = a, g$, due to limited computational resources. A practical approach is to stop iteration when the following criterion is satisfied:

$$\begin{aligned} \|J_g^{t+1}(l) \to J_g^t(l)\| \le \epsilon \\ \|J_a^{t+1}(\tau) \to J_a^t(\tau)\| \le \epsilon \end{aligned} \tag{19}$$

leading to ϵ-optimal value functions $J_\star^\epsilon$, $\star = a, g$, with tolerance error ϵ. The associated ϵ-optimal control policies are u_a^ϵ for UAVs and $u_g^{[N],\epsilon}$ for UGVs, respectively.

Suppose that the prediction horizon N_a has been selected such that the optimal control $\overrightarrow{\boldsymbol{u}_a}^*(0)$ at time $h = 0$ satisfies $e_{a,i}(N_a - N + 1), \ldots, e_{a,i}(N_a)$ being in the neighbor of the origin, and the optimal control $\overrightarrow{\boldsymbol{u}_g}^{[N],*}(0)$ at time $k = 0$ satisfies $e_{g,i}^{[N]}(N_g)$ being in the neighbor of the origin. Assume that the terminal condition (36) deferred in Appendix B holds. Then the following theorem is stated.

Theorem 3 (Closed-loop stability at fast/slow time scales): Under Assumptions 1 and 2, the closed-loop system under Algortihm 1 are such that:

1) $e_a(h)$ and $\boldsymbol{u}_a(h)$ converge to the origin asymptotically at the fast time scale h.
2) $e_g^{[N]}(k)$ and $\boldsymbol{u}_g^{[N]}(k)$ converge to the origin asymptotically at the slow time scale k.

Proof: Please refer to Appendix D. ■

IV. Multirate Distributed Actor-Critic Implementation

Note that, Algorithm 1 is not ready for online implementation due to the computational complexity in (17b) and (18b). Thereby, this section offers an efficient distributed actor-critic implementation of the proposed multirate distributed RHRL framework. We first present the distributed actor-critic structure at fast and slow time scales. Then, we give the summarized multirate distributed actor-critic learning algorithm for online implementation. Note that our approach's policy learning procedures are performed incrementally in each prediction interval, facilitating fast online learning abilities.

A. Distributed Actor-Critic Learning at Slow Time Scale

The implementation of the UGV formation control consists of several actor-critic pairs, each pair designed for each UGV to approximate the value function and the control policy at the slow time scale. We first introduce the structure of the critic and actor for the UGVs.

Critic learning: For the ith UGV, define the costate as $\lambda_{g,i}(l) = \partial J_{g,i}(l)/\partial \bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l)$ for $l \in [k,\ k + N_g]$. The critic network is constructed to learn the costate, that is,

$$\hat{\lambda}_{g,i}(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l)) = W_{g,i}^\top \sigma_{g,i}(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l), l), \quad i \in \mathbb{N}_1^{M_g} \tag{20}$$

for all $l \in [k, k + N_g - 1]$, where $W_{g,i} \in \mathbb{R}^{N_{wg,i} \times \bar{n}_{g,i}}$ is the weighting matrix to be updated during learning, $\sigma_{g,i} \in \mathbb{R}^{N_{wg,i}}$ is a vector composed of basis functions.

As the optimal costate $\lambda_{g,i}^*(l)$ is unavailable, we define $\lambda_{g,i}^d(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l))$ as the desired value of the output of the critic

$$\lambda_{g,i}^d(l) = 2\bar{Q}_i \bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l) + \sum_{j \in \bar{\mathcal{N}}_{g,i}} \left(\frac{\partial f_{g,j}(\bar{e}_{\mathcal{N}_{g,j}}^{[N]}(l))}{\partial \bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l)} \right)^\top \hat{\lambda}_{g,i}^{[j]}(l+1) \tag{21}$$

for $l \in [k, k + N_g - 1]$, where $\hat{\lambda}_{g,i}^{[j]}$ is the associated entries of $\hat{\lambda}_{g,i}$ of the jth UAV or UGV connected to the ith UGV.

Minimizing the cost $\delta_{g,i}(l) = \|\epsilon_{g,i}(l)\|^2 = \|\lambda_{g,i}^d(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l)) - \hat{\lambda}_{g,i}(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l))\|^2$ leads to the update rule of $W_{g,i}$ as

$$W_{g,i}(l+1) = W_{g,i}(l) - \gamma_{g,i} \frac{\partial \delta_{g,i}(l)}{\partial W_{g,i}(l)} \tag{22}$$

where $\gamma_{g,i} \in \mathbb{R}^+$ is the local learning rate.

Actor learning: Likewise, for ith UGV, the actor network is constructed as follows:

$$\hat{u}_{g,i}^{[N]}(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l)) = V_{g,i}^\top \phi_{a,i}(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l), l) \tag{23}$$

where $V_{g,i} \in \mathbb{R}^{N_{vg,i} \times 2}$ is the weighting matrix, $\phi_{g,i} \in \mathbb{R}^{N_{vg,i}}$ is a vector composed of basis functions. In view of the first-order optimality condition of the HJB (15), a desired target of $u_{g,i}^{[N]}$ is defined as

$$u_{g,i}^{[N],d}(l) := -\frac{1}{2} R_{g,i}^{-1} \sum_{j \in \bar{\mathcal{N}}_{g,i}} g_{g,i}^\top(e_{g,i}^{[N]}(l)) \hat{\lambda}_{g,j}^{[i]}(l+1) \tag{24}$$

$l \in [k, k + N_g - 1]$. Let $\varepsilon_{g,i}(l) = u_{g,i}^{[N],d}(l) - \hat{u}_{g,i}^{[N]}(l)$. Minimizing $\kappa_{g,i}(l) = \|\varepsilon_{g,i}(l)\|^2$ leads to the update rule of $V_{g,i}$ as

$$V_{g,i}(l+1) = V_{g,i}(l) - \eta_{g,i} \frac{\partial \kappa_{g,i}(l)}{\partial V_{g,i}(l)} \tag{25}$$

where $\eta_{g,i} \in \mathbb{R}^+$ is the local learning rate.

B. Distributed Actor-Critic Learning at Fast Time Scale

Analogously, implementing the UAV formation control consists of several actor-critic pairs working at the fast time scale.

Critic learning: For the ith UAV, define the costate as $\lambda_{a,i}(\tau) = \partial J_{a,i}(\tau)/\partial \bar{e}_{\mathcal{N}_{a,i}}(\tau)$ for $\tau \in [h, h + N_a - q]$. The critic network is constructed to learn the costate, that is,

$$\hat{\lambda}_{a,i}(\bar{e}_{\mathcal{N}_{a,i}}(\tau)) = W_{a,i}^\top \sigma_{a,i}(\bar{e}_{\mathcal{N}_{a,i}}(\tau), \tau), \quad i \in \mathbb{N}_1^{M_a} \tag{26}$$

for all $\tau \in [h, h + N_a - q - 1]$, where $W_{a,i} \in \mathbb{R}^{N_{wa,i} \times \bar{n}_{a,i}}$ is the weighting matrix to be updated during learning, $\sigma_{a,i} \in \mathbb{R}^{N_{wa,i}}$ is a vector composed of basis functions.

As the optimal costate $\lambda_{a,i}^*(\bar{e}_{\mathcal{N}_{a,i}}(\tau))$ is unavailable, the desired value of $\hat{\lambda}_{a,i}(\bar{e}_{\mathcal{N}_{a,i}}(\tau))$ is defined as follows:

$$\begin{aligned} \lambda_{a,i}^d(\tau) = {} & 2\bar{Q}_i \bar{e}_{\mathcal{N}_{a,i}}(\tau) \\ & + \sum_{j \in \bar{\mathcal{N}}_{a,i}} \left(\frac{\partial f_{a,j}(\bar{e}_{\mathcal{N}_{a,j}}(\tau))}{\partial \bar{e}_{\mathcal{N}_{a,j}}(\tau)} \right)^\top \hat{\lambda}_{a,i}^{[j]}(\tau+1) \end{aligned} \tag{27}$$

for $\tau \in [h, h + N_a - q - 1]$, where $\hat{\lambda}_{a,i}^{[j]}(\bar{e}_{\mathcal{N}_{a,i}})$ is the associated entries of $\hat{\lambda}_{a,i}(\bar{e}_{\mathcal{N}_{a,i}})$ corresponding to jth UGV or UAV connected to the ith UAV.

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2: Multirate Distributed Actor-Critic Implementation in Each Time Interval $[kN,\ kN+N-1]$.

Procedure 1: Policy learning at $h = kN$.

Require:

Initialize $W_{*,i}$ and $V_{*,i}$, $* = g, a$, with uniformly distributed random matrices;
Set $\bar{\epsilon} > 0$, Err $\geq \bar{\epsilon}$, $t = 0$, $t_{\max}$;
Set $W^0_{*,i} = W_{*,i}$ and $V^0_{*,i} = V_{*,i}$, $* = g, a$;
while Err $\geq \bar{\epsilon} \vee t \leq t_{\max}$ **do**
  for $\tau = kN, \cdots, kN + N_a - 1$ **do**
    if mode$(\tau, N) = 0$ **then**
      $l = \lfloor \tau / N \rfloor$;
      Calculate $e^{[N]}_{g,i}(l+1)$ with $\hat{u}^{[N]}_{g,i}(\bar{e}^{[N]}_{\mathcal{N}_{g,i}}(l))$ using (8), $\forall\, i \in \mathbb{N}_1^{M_g}$;
      Compute $\hat{\lambda}_{g,i}(l)$ with $\bar{e}^{[N]}_{\mathcal{N}_{g,i}}(l)$ and $\hat{\lambda}_{g,i}(l+1)$ using the predicted state $\bar{e}^{[N]}_{\mathcal{N}_{g,i}}(l+1)$ with (20), $\forall\, i \in \mathbb{N}_1^{M_g}$;
      Derive $\lambda^d_{g,i}(l)$ with (21) and $u^{[N],d}_{g,i}(l)$ with (24), $\forall\, i \in \mathbb{N}_1^{M_g}$;
      Update $W_{g,i}$ with (22) and $V_{g,i}$ with (25), $\forall\, i \in \mathbb{N}_1^{M_g}$;
    end if
  end for
  Set $\hat{u}_{g,i}(\tau) = \lfloor \hat{u}^{[N]}_{g,i}(\bar{e}^{[N]}_{\mathcal{N}_{g,i}}(l)) \rfloor$ and apply to the model derived with (7) and (2);
  for $\tau = kN, \cdots, kN + N_a - 1$ **do**
    Calculate $e_{a,i}(\tau+1)$ with $\hat{u}_{a,i}(\bar{e}_{\mathcal{N}_{a,i}}(\tau))$ using (6), $\forall\, i \in \mathbb{N}_1^{M_a}$;
    Compute $\hat{\lambda}_{a,i}(\tau)$ with $\bar{e}_{\mathcal{N}_{a,i}}(\tau)$ and $\hat{\lambda}_{a,i}(\tau+1)$ using the predicted state $\bar{e}_{\mathcal{N}_{a,i}}(\tau+1)$ with(26), $\forall\, i \in \mathbb{N}_1^{M_a}$;
    Derive $\lambda^d_{a,i}(\tau)$ with (27) and $u^d_{a,i}(\tau)$ with (30), $\forall\, i \in \mathbb{N}_1^{M_a}$;
    Update $W_{a,i}$ with (28) and $V_{a,i}$ with (31), $\forall\, i \in \mathbb{N}_1^{M_a}$;
  end for
  Set $W^{t+1}_{*,i} = W_{*,i}$ and $W^{t+1}_{*,i} = W_{*,i}$;
  Compute

$$\text{Err} = \sum_{i=1}^{M_*} \|W^{t+1}_{*,i} - W^t_{*,i}\| + \|V^{t+1}_{*,i} - V^t_{*,i}\|;$$

  $t \leftarrow t+1$;
end while
Update $e^{[N]}_{g,i}(k+1)$ with control (23) and update $e_{a,i}(kN+1)$ with control (29).
Set $u_{g,i}(\tau) = \hat{u}^{[N]}_{g,i}(\lfloor \tau / kN \rfloor)$, and update the state predictions related to the UGV using (3b) and (7).

Procedure 2: Policy learning at $h \in [kN+1, kN+N-1]$.

Require: $t = 0$, $t_{\max}$;

while $t \leq t_{\max}$ **do**
  for $\tau = h, \cdots, h + N_a - q - 1$ **do**
    Repeat steps 16-19 for the UAV team;
  end for
  $t \leftarrow t+1$;
end while

Minimizing the cost $\delta_{a,i}(\tau) = \|\epsilon_{a,i}(\tau)\|^2 = \|\lambda^d_{a,i}\ (\bar{e}_{\mathcal{N}_{a,i}}(\tau)) - \hat{\lambda}_{a,i}\ (\bar{e}_{\mathcal{N}_{a,i}}\ (\tau))\|^2$ leads to the update rule of $W_{a,i}$ as

$$W_{a,i}(\tau+1) = W_{a,i}(\tau) - \gamma_{a,i} \frac{\partial \delta_{a,i}(\tau)}{\partial W_{a,i}(\tau)} \tag{28}$$

where $\gamma_{a,i} \in \mathbb{R}^+$ is the local learning rate.

Actor learning: For the ith UAV, we construct the actor network as

$$\hat{u}_{a,i}(\bar{e}_{\mathcal{N}_{a,i}}(\tau)) = V^\top_{a,i} \phi_{a,i}(\bar{e}_{\mathcal{N}_{a,i}}(\tau), \tau) \tag{29}$$

where $V_{a,i} \in \mathbb{R}^{N_{va,i} \times 3}$ is the weighting matrix, $\phi_{a,i} \in \mathbb{R}^{N_{va,i}}$ is a vector composed of basis functions. In view of the first-order optimality condition of the HJB (16), we define a desired target of $u_{a,i}$ as

$$u^d_{a,i}(\tau) := -\frac{1}{2} R^{-1}_{g,i} \sum_{j \in \bar{\mathcal{N}}_{g,i}} g^\top_{a,i}(e_{a,i}(\tau)) \hat{\lambda}^{[i]}_{a,j}(\tau+1) \tag{30}$$

$\tau \in [h, h+N_a-q-1]$. Letting $\kappa_{a,i}(\tau) = u^d_{a,i}(\tau) - \hat{u}_{a,i}(\tau)$, at each time $\tau \in [h, h+N_a-1]$, each robot i minimizes $\kappa_{a,i}(\tau) = \|\varepsilon_{a,i}(\tau)\|^2$, leading to the update rule of $V_{a,i}$ as

$$V_{a,i}(\tau+1) = V_{a,i}(\tau) - \eta_{a,i} \frac{\partial \kappa_{a,i}(\tau)}{\partial V_{a,i}(\tau)} \tag{31}$$

where $\eta_{a,i} \in \mathbb{R}^+$ is the local learning rate.

C. Summary of the Multirate Actor-Critic Implementation

The implementation of the multirate distributed actor-critic learning is summarized as follows. At each time interval $[kN, kN+N-1]$, the policy learning is divided into two procedures. First, at the time instant $h = kN$, the actor-critic networks of the UGVs and UAVs are updated in an asynchronous way on different time scales with a fixed prediction horizon N_a, while only the actor-critic networks associated with the UAV team are updated for learning the control policies in the subsequent time interval $h \in [kN+1, kN+N-1]$, but with a shrinking prediction horizon. The above procedures are repeatedly solved in the next time interval $[kN+1,\ kN+N]$, and the detailed implementation is described in Algorithm 2.

V. Simulation Results

This section validates the effectiveness of our proposed multirate distributed RHRL approach for UAV–UGV formation in various scenarios, including different formation goals and different time scales. In the simulation, the UAVs and UGVs move cooperatively with predefined formation shapes to perform a cooperative search task. Therein, several UAVs work at a fast time scale to quickly collect mission-critical environmental information in one specific formation, while a number of UGVs operating in a slow time scale are aware of the perception information in the forward hazardous region in advance and follow smooth formation shapes at a slower speed. The following fully demonstrates the effectiveness of our proposed approach under different time scale settings and formation tasks and, showing the superiority in control performance to the distributed RL method [39] working at single time scales.

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

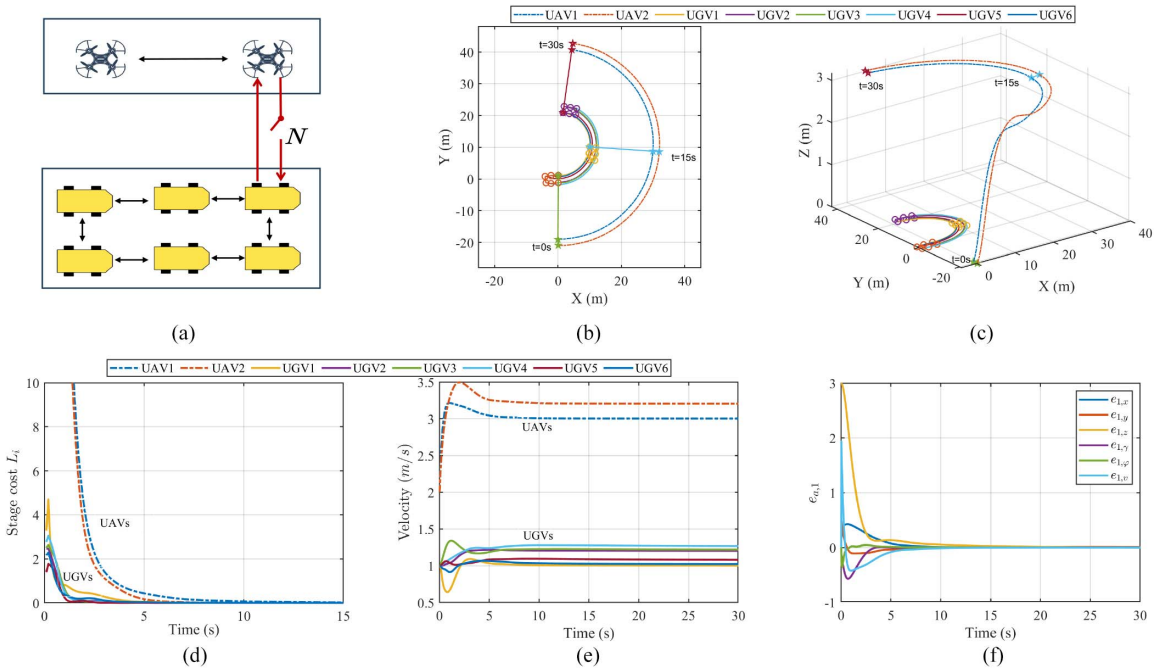


Fig. 3. Multirate UAV–UGV cooperation in a circular formation scenario. (a) Communication topology of the UAV–UGV system. (b) UAV–UGV formation paths of UAVs–UGVs mapped in the 2-D plane. (c) UAV–UGV formation paths in the 3-D space. (d) Variations of the stage costs of UAVs and UGVs. (e) Velocities of UAVs and UGVs. (f) Formation errors of the first UAV.

A. Multirate UAV–UGV Cooperation in Circular Formation

We start with a circular formation example to verify the effectiveness of our approach. In the simulation, the UAVs and UGVs are coordinated in the horizontal plane to form a circular formation. The communication graph of the UAVs and UGVs is displayed in Fig. 3. The UAVs worked at a basic time scale with the sampling interval $\Delta t = 0.02\,s$ to form a circular shape with larger radiuses and speeds, while the UGVs worked at the slower time scale with $N = 5$ (corresponding to a sampling interval of $0.1\,s$) formed circular shapes with lower radius and speeds. The simulation results are shown in Fig. 3, while the simulation setup and the choices of detailed parameters are omitted here and deferred in Appendix E. As shown in Panels B and C of Fig. 3, the UAVs reacted promptly to form the circular formation in the horizontal plane despite the vertical direction being controlled independently. The stage costs of the UAVs and UGVs converge asymptotically to the origin (see Panel D in Fig. 3), which implies that the formation errors and the control inputs of UAVs and UGVs converge to the origin in view of the definition of the stage costs (see also Panel F). Note that the UAVs during formation generation must account for the external influences by the variations of the pitch angle and attitude, which changed rapidly during taking off, causing a delay in convergence speed. Also, as illustrated in Fig. 3, the velocities of the UAVs are almost three times larger than that of the UGVs. To verify the learning ability of our approach, we perform multiple runs using the learned weights from the last episode to initialize for the subsequent episode. The results in Fig. 4 show that our approach enables high learning efficiency under the receding horizon learning mechanism. The verification of our approach to more practical scenarios is considered in the next sections.

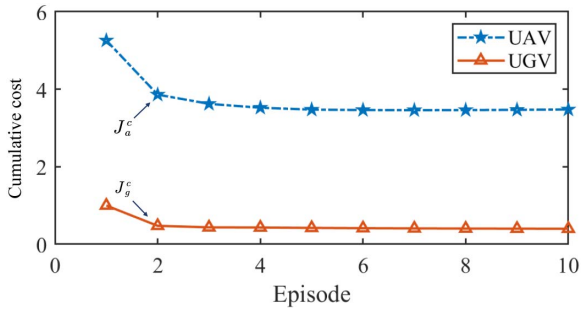


Fig. 4. Learned cumulative cost with the running episode to go, where $J_a^c = \sum_{i=1}^{M_a}\sum_{h=1}^{N_{\text{sim}}} L_{a,i}(h)$, $J_g^c = \sum_{i=1}^{M_g}\sum_{k=1}^{N_{\text{sim}}/N} L_{g,i}(k)$, $N_{\text{sim}} = 1500$.

B. Cooperative Search Task With Different Formation Goals

This section considers a more practical UAV–UGV formation control scenario for cooperative search tasks. In this scenario, the UAVs and UGVs could have their own formation goals, but the geometric center of the UGV formation is transmitted to UAVs for flexible distance coordination between the UAVs and UGVs (see Panel A of Fig. 5). In this way, UAVs can form many different formation patterns in a large area for various applications. Specifically, UGVs work at the slow time scale associated with $\Delta t = 0.1\,s$ to form a rectangular formation, while the UAVs are operated at the fast time scale ($\Delta t = 0.02\,s$) to form a distinct formation goal for completing potential formation tasks. As shown in Panels A and B of Fig. 5, the moving

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

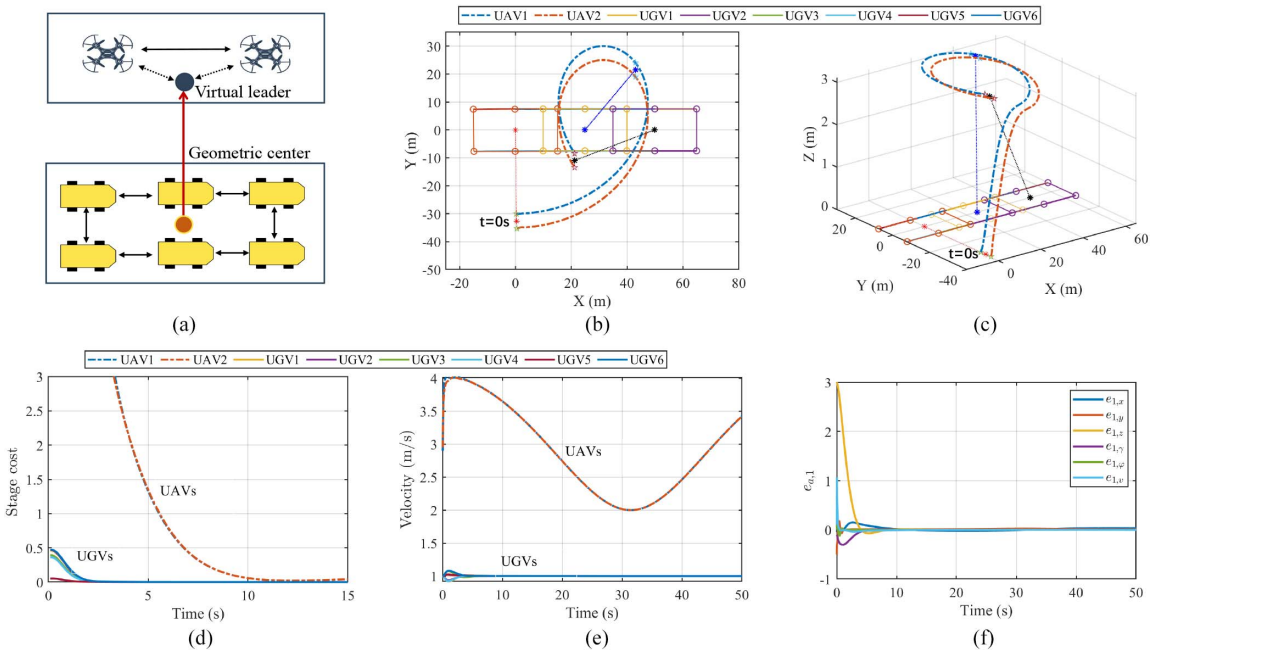


Fig. 5. Multirate UAV–UGV cooperation in a cooperative search scenario where UGVs retain the rectangular formation, and the UAVs keep the circular formation. (a) Communication topology of the UAV–UGV system. (b) UAV–UGV formation paths of UAVs and UGVs mapped in the 2-D plane. (c) UAV–UGV formation paths in the 3-D space. (d) Variations of the stage costs of UAVs and UGVs. (e) Velocities of UAVs and UGVs. (f) Formation errors of the first UAV.

trajectory of the UAVs' virtual leader was a circular curve with a dynamic center that was given by the geometric center of the UGVs.

Under the detailed parameter settings presented in Appendix E, Algorithm 2 was implemented to learn the optimal formation control strategies for the UAV–UGV team. The air-ground system's associated 3-D and 2-D trajectories are presented in Panels B and C of Fig. 5, demonstrating that the UAVs and UGVs working at different time scales cooperatively accomplish the search task in the large area. As presented in Panel D of Fig. 5, the stage costs of the UAVs and UGVs converge asymptotically to the origin; this concludes that the formation errors and the control inputs of UAVs and UGVs all converge to the origin. The trajectories of formation errors of the UAV and UGV teams presented in Panel F of Fig. 5 all converge to the origin, which validates the effectiveness of our proposed approach to drive UAVs and UGVs to keep the predetermined formation for accomplishing given tasks. Moreover, the simulation results in Fig. 6 have verified the effectiveness of our approach under the different time scale separation settings with $\Delta t = 0.05\,s$ and $N = 2$.

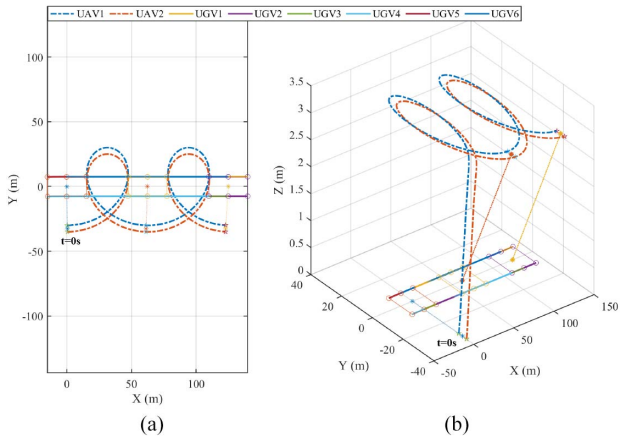


Fig. 6. Multirate UAV–UGV cooperation in a cooperative search scenario with $\Delta t = 0.05\,s$ and $N = 2$. (a) UAV–UGV formation paths mapped in the 2-D plane. (b) UAV–UGV formation paths in the 3-D space.

C. Scalability Validation With Expanded Unmanned Systems

This section validates the scalability of our proposed framework under a coordinated mission including 6 UAVs (sampling interval: 0.02 s) and 12 UGVs (sampling interval: 0.1 s). UAVs maintain a circular formation, while UGVs form a rectangular formation. The UAV formation centre dynamically tracks the geometric centre of the UGVs. Based on the parameter settings in Appendix E, the formation control performance of the proposed framework is shown in Fig. 7. The 2-D and 3-D trajectories of the UAV–UGV team are presented in Panels A and B. The formation error curves of 6 UAVs and 12 UGVs are shown in Panels C and D. The presented results demonstrate that coordination errors asymptotically converge to zero, validating the scalability of the proposed method towards heterogeneous unmanned systems.

D. Robustness Validation with Expanded Unmanned Systems

Following the simulation and parameter settings in Section V-C, this section further validates the robustness of our

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

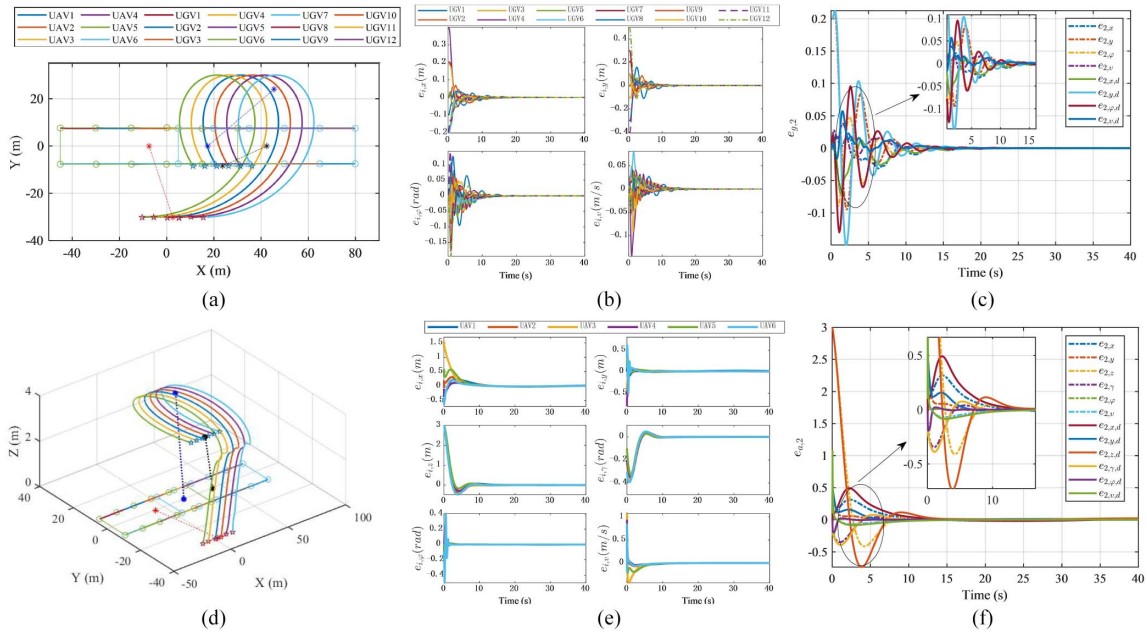


Fig. 7. Scalability and robustness of the proposed framework under a coordinated mission including 6 UAVs and 12 UGVs. (a) UAV–UGV formation paths mapped in the 2-D plane. (b) UAV–UGV formation paths in the 3-D space. (c) Formation errors of 12 UGVs. (d) Formation errors of 6 UAVs. (e) Formation error of the second UGV under disturbance. (f) Formation error of the second UAV under disturbance.

proposed framework under the additionally introduced disturbance provided in Appendix V-C. The asymptotically converged formation errors presented in Panels E and F of Fig. 7 validate that the proposed formation could drive the UAV–UGV team to complete the given coordination task even under disturbances.

E. Comparisons With Algorithms Under Single Time Scales

This section conducts comparative simulations to show the superiority of our proposed multirate distributed RHRL approach (see the first line in Table I) over the state-of-the-art (SOTA) methods where UAVs and UGVs worked at single time scales. The adopted scenario is identical to that in Section V-B. The detailed parameter settings to conduct the comparative simulations are referred to in Appendix E. The results presented in Table I show that RHRL outperforms other SOTA methods regarding both performance cost and communication overhead. Specifically, compared to the fast-timescale-only methods (both learning and communicating at h), our RHRL framework reduces communication load by 80% for slow UGVs, and reduces performance degradation caused by intergroup communication on UAVs. Compared to the slow-timescale-only methods (both learning and communicating at k), RHRL better matches UAV dynamics, reduces intergroup state errors, and improves coordination in UAV–UGV systems. The EBLT method [41], when used at slow timescales, lacks adaptability and fails to meet real-time UAV control needs. In contrast, both our RHRL and ADPF [42] are learning-based and maintain control effectiveness even with large sampling periods. The proposed method further enhances online learning and adaptability via a receding horizon scheme, outperforming ADPF. Overall, our proposed

TABLE I
NUMERICAL COMPARISONS WITH ALGORITHMS WORKING AT SINGLE TIME SCALES

Methods	Learning at		Communicating at		Average cost		
	UGV	UAV	UGV	UAV	J_g^c	J_a^c	J_t^c
Ours	$\forall k$	$\forall h$	$\forall k$	$\forall h$	**0.38**	**15.80**	**16.18**
SRL [39] (fast)	$\forall h$	$\forall h$	$\forall h$	$\forall h$	0.59	16.98	17.57
SRL [39] (slow)	$\forall k$	$\forall k$	$\forall k$	$\forall k$	0.51	29.50	30.01
ADPF [42] (fast)	$\forall h$	$\forall h$	$\forall h$	$\forall h$	0.76	21.62	22.38
ADPF [42] (slow)	$\forall k$	$\forall k$	$\forall k$	$\forall k$	0.63	45.61	46.24
EBLT [41] (fast)	$\forall h$	$\forall h$	$\forall h$	$\forall h$	1.34	35.51	36.85
EBLT [41] (slow)	$\forall k$	$\forall k$	$\forall k$	$\forall k$	–	–	–

Note: (1) $\bar{J}_g^c = 1/N_s \sum_{j=1}^{N_s} \sum_{i=1}^{M_a} \sum_{k=1}^{N_{\text{sim}}/N} L_{g,i}^{[j]}(k)$, where N_s is the success number in 100 tests. (2)] $\bar{J}_a^c = 1/N_s \sum_{j=1}^{N_s} \sum_{i=1}^{M_g} \sum_{h=1}^{N_{\text{sim}}} L_{a,i}^{[j]}(h)$. $\bar{J}_t^c = \bar{J}_g^c + \bar{J}_a^c$.

RHRL framework reduces performance cost by over 7.9% compared to fast-timescale-only methods and over 46% compared to slow-timescale-only methods. By assigning UAVs and UGVs to different time scales and leveraging dual-timescale state information, our method outperforms the SOTA methods.

F. Validation in Gazebo

This section validates our method through formation control of a UAV–UGV team (12 UAVs, 6 UGVs) in Gazebo, simulated in Python using the XTdrone platform [43]. The results presented in Fig. 8 further validate the scalability and robustness of our framework. In particular, Fig. 8(a)–(f) provide the snapshots of critical mission states. The converged formation errors of 6 UGVs and 12 UAVs, shown in Fig. 8(g) and (h) respectively,

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

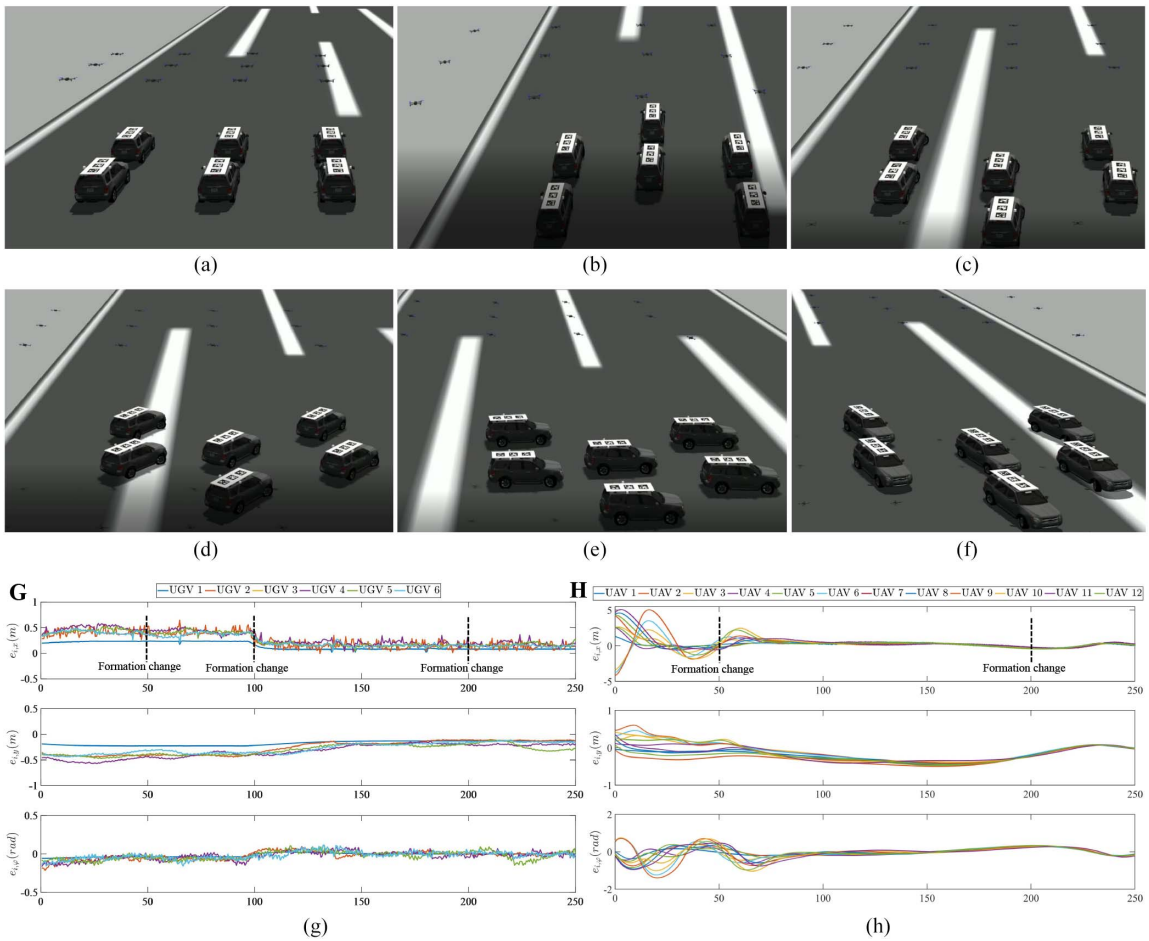


Fig. 8. Experimental validation of the proposed framework under a coordinated mission including 12 UAVs and 6 UGVs. (a)–(f) Snapshots of critical mission states. (g) Formation error of the UGVs. (h) Formation error of the UAVs. The fluctuations in formation error seen in (g) and (h) result from the formation change illustrated in snapshots (a)–(f).

demonstrate that our method could drive the UAV–UGV team to maintain a predefined formation to complete given tasks.

VI. Conclusion

In this article, we have proposed a multirate distributed RHRL approach for optimal UAV–UGV formation control. In view of the main characteristics, our approach utilizes a time-scale separation technique to formulate the UAV–UGV formation problem as a multirate control problem, which is solved by two distributed RHRL algorithms working on different time scales. One is designed at a faster time scale for realizing optimal UAV formation control, while another is designed at a slower time scale for optimal UGV formation control. Compared with multirate control approaches based on MPC in the process control domain, our approach learns explicit closed-loop control policies rather than open-loop control sequences. Consequently, the learned policies can be directly deployed to different control scenarios without fine-tuning. The proposed approach is potentially useful in many robotic applications, such as the coordination of UAVs and unmanned surface vehicles for search and rescue in the sea area, and can be extended to a hierarchical multirate control scenario [6]. One of our limitations lies in the use of quadratic cost functions in the performance index. Future work will extend our approach to cope with nonquadratic cost functions and to more general swarm control scenarios with collision avoidance capabilities.

Appendix

A. Terminal Penalty Design

In receding horizon control theory, the design of terminal penalty matrices and terminal constraints is crucial to derive the closed-loop stability. The core idea is to enforce the terminal state in a positively invariant set such that the terminal cost serves as a Lyapunov function that maintains the future state staying in the invariant set under a computable stabilizing control law. As shown in [44], [45], the terminal constraint can be removed in the controller design. The terminal state will eventually approach the neighbor of the origin as long as the prediction horizon is large enough. Hence, for the sake of simplicity, we focus only on the design of the penalty matrices at slow and fast time scales.

In particular, inspired by the promising work [40] that designs the penalty matrix for a distributed MPC approach of

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

linear systems, we move a step further by extending the design to the multirate control of nonlinear systems.

First, for the optimization problem (9), letting $\bar{Q}_{g,i} = \text{blkdiag}\{Q_{g,i}, Q_{a\to g,i}\}$, we choose $P_{g,i}$ as the solution to the following Lyapunov equation:

$$F_{g,i}^\top P_{g,i} F_{g,i} - \bar{P}_{g,i} = -\beta_{g,i}\left(\bar{Q}_{g,i} + K_{\mathcal{N}_{g,i}}^\top R_{g,i} K_{\mathcal{N}_{g,i}}\right) + \Gamma_{\mathcal{N}_{g,i}} \tag{32}$$

$\forall i \in \mathbb{N}_1^{M_g}$, where the scalar $\beta_{g,i} > 1$, $\bar{P}_{g,i} := E_{g,i} T_{g,i}^\top P_{g,i} T_{g,i} E_{g,i}^\top$ lifts $P_{g,i}$ to the space of neighboring states belonging in $\mathbb{R}^{\bar{n}_{g,i}}$, $T_{g,i} \in \{0,1\}^{4\times n_g}$ and $E_{g,i} \in \{0,1\}^{\bar{n}_{g,i}\times n_g}$ are selective matrices such that $e_{g,i}^{[N]} = T_{g,i} e_g^{[N]}$, $\bar{e}_{\mathcal{N}_{g,i}}^{[N]} = W_{g,i} e_g^{[N]}$ with $e_g^{[N]} \in \mathbb{R}^{n_g}$, $n_g = 4M_g$; and $\sum_{i=1}^{M_g} E_{g,i}^\top \Gamma_{\mathcal{N}_{g,i}} E_{g,i} \le 0$; $F_{g,i} = A_{\mathcal{N}_{g,i}} + B_{g,i} K_{\mathcal{N}_{g,i}}$, $A_{\mathcal{N}_{g,i}} \in \mathbb{R}^{4\times n_{\mathcal{N}_{g,i}}}$ and $B_{g,i} \in \mathbb{R}^{4\times 2}$ are the model matrices of the linearized model of (8) around the origin, that is,

$$\begin{aligned} &e_{g,i}^{[N]}(k+1) \\ &= A_{\mathcal{N}_{g,i}} \bar{e}_{\mathcal{N}_{g,i}}^{[N]}(k) + B_{g,i} u_{g,i}^{[N]}(k) + \text{Er}_{g,i}(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(k), u_{g,i}^{[N]}(k)) \end{aligned} \tag{33}$$

$\text{Er}_{g,i}$ is the linearization error and, $\lim_{\bar{e}_{\mathcal{N}_{g,i}}^{[N]}, u_{g,i}^{[N]}\to 0} \text{Er}_{g,i}(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}, u_{g,i}^{[N]})/(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}, u_{g,i}^{[N]}) \to 0$; the matrix $K_{\mathcal{N}_{g,i}} \in \mathbb{R}^{2\times n_{\mathcal{N}_{g,i}}}$ satisfying $u_g^{[N]} = \text{col}_{i\in\mathbb{N}_1^{M_g}}(K_{\mathcal{N}_{g,i}} \bar{e}_{\mathcal{N}_{g,i}}^{[N]})$ is a stabilizing control policy.

Analogously, for the optimization problem (12), letting $\bar{Q}_{a,i} = \text{blkdiag}\{Q_{a,i}, Q_{g\to a,i}\}$, $\tilde{Q}_{a,i} = \bar{Q}_{a,i} + K_{\mathcal{N}_{a,i}}^\top R_{a,i} K_{\mathcal{N}_{a,i}}$, we choose $P_{a,i}$ as the solution to the following Lyapunov equation:

$$\begin{aligned} &F_{a,i}^\top P_{a,i} F_{a,i} + \sum_{\tau=0}^{N-1} (F_{a,i}^\tau)^\top \tilde{Q}_{a,i} F_{a,i}^\tau - \bar{P}_{a,i} \\ &= -\beta_{a,i} \tilde{Q}_{a,i} + \Gamma_{\mathcal{N}_{a,i}} \end{aligned} \tag{34}$$

$\forall i \in \mathbb{N}_1^{M_a}$, where the scalar $\beta_{a,i} > 1$, $\bar{P}_{a,i} := E_{a,i} T_{a,i}^\top P_{a,i} T_{a,i} E_{a,i}^\top$ lifts $P_{a,i}$ to the space of neighboring states belonging in $\mathbb{R}^{\bar{n}_{a,i}}$, $T_{a,i} \in \{0,1\}^{6\times n_a}$ and $E_{a,i} \in \{0,1\}^{\bar{n}_{a,i}\times n_a}$ are selective matrices such that $e_{a,i} = T_{a,i} e_a$, $e_{\mathcal{N}_{a,i}} = E_{a,i} e_a$ with $e_a \in \mathbb{R}^{n_a}$, $n_a = 6M_a$; and $\sum_{i=1}^{M_a} E_{a,i}^\top \Gamma_{\mathcal{N}_{a,i}} E_{a,i} \le 0$; $F_{a,i} = A_{\mathcal{N}_{a,i}} + B_{a,i} K_{\mathcal{N}_{a,i}}$, $A_{\mathcal{N}_{a,i}} \in \mathbb{R}^{6\times n_{\mathcal{N}_{a,i}}}$ and $B_{a,i} \in \mathbb{R}^{6\times 3}$ are the model matrices of the linearized model of (6) around the origin, that is,

$$\begin{aligned} &e_{a,i}(k+1) \\ &= A_{\mathcal{N}_{a,i}} \bar{e}_{\mathcal{N}_{a,i}}(k) + B_{a,i} u_{a,i}(k) + \text{Er}_{a,i}(\bar{e}_{\mathcal{N}_{a,i}}(k), u_{a,i}(k)) \end{aligned} \tag{35}$$

$\text{Er}_{a,i}$ is the linearization error and $\lim_{\bar{e}_{\mathcal{N}_{a,i}}, u_{a,i}\to 0} \text{Er}_{a,i}(\bar{e}_{\mathcal{N}_{a,i}}, u_{a,i})/(\bar{e}_{\mathcal{N}_{a,i}}, u_{a,i}) \to 0$; the matrix $K_{\mathcal{N}_{a,i}} \in \mathbb{R}^{3\times n_{\mathcal{N}_{a,i}}}$ satisfying $u_a = \text{col}_{i\in\mathbb{N}_1^{M_a}}(K_{\mathcal{N}_{a,i}} \bar{e}_{\mathcal{N}_{a,i}})$ is a stabilizing control policy.

B. Terminal Conditions

To derive the closed-loop stability, we also need the following conditions. Let $\text{Er}_{a,i}(\bar{e}_{\mathcal{N}_{a,i}}) = \text{Er}_{a,i}(\bar{e}_{\mathcal{N}_{a,i}}, K_{\mathcal{N}_{a,i}} \bar{e}_{\mathcal{N}_{a,i}})$ and $\text{Er}_{g,i}(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}) = \text{Er}_{g,i}(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}, K_{\mathcal{N}_{g,i}} \bar{e}_{\mathcal{N}_{g,i}}^{[N]})$. Denote $S_{a,i} = \sup \|\text{Er}_{a,i}(\bar{e}_{\mathcal{N}_{a,i}})\|/\|\bar{e}_{\mathcal{N}_{a,i}}\| \to 0$ and $S_{g,i} = \sup \|\text{Er}_{g,i}(\bar{e}_{\mathcal{N}_{g,i}}^{[N]})\|/\|\bar{e}_{\mathcal{N}_{g,i}}^{[N]}\| \to 0$ in a small neighbor of the origin, and let $\beta_\star$ and $P_\star$ for $\star = a, g$ be selected such that

$$\begin{aligned} \|P_{g,i}\| S_{g,i}^2 + 2\|P_{g,i} F_{g,i}\| S_{g,i} &< (\beta_{g,i} - 1)\lambda_{\min}(\tilde{Q}_{g,i}) \\ \Pi(\tilde{Q}_{a,i}, S_{a,i}) + \Pi(P_{a,i}, S_{a,i}) &< (\beta_{a,i} - 1)\lambda_{\min}(\tilde{Q}_{a,i}) \end{aligned} \tag{36}$$

where $\tilde{Q}_{\star,i} = \bar{Q}_{\star,i} + K_{\mathcal{N}_{\star,i}}^\top R_{\star,i} K_{\mathcal{N}_{\star,i}}$, and

$$\begin{aligned} &\Pi(P_{a,i}, S_{a,i}) \\ &= \sum_{j=0}^{N-1} (F_{a,i}^j)^\top P_{a,i} F_{a,i}^j S_{a,i}^2 + 2(F_{a,i}^j)^\top P_{a,i} F_{a,i}^{j+1} S_{a,i} \end{aligned}$$

$$\begin{aligned} &\Pi(\tilde{Q}_{a,i}, S_{a,i}) \\ &= \sum_{\tau=0}^{N-1} \sum_{j=0}^{\tau-1} (F_{a,i}^j)^\top \tilde{Q}_{a,i} F_{a,i}^j S_{a,i}^2 + 2(F_{a,i}^j)^\top \tilde{Q}_{a,i} F_{a,i}^{j+1} S_{a,i}). \end{aligned}$$

C. Proof to Theorem 1

Proof: Claim (i): First, by collecting (18a) for $i \in \mathbb{N}_1^{M_g}$, one can write the value update of the UGV at the slow time scale in a collective form as follows:

$$J_g^{t+1}(l) = L_g(l) + J_g^t(l+1),\ l = \lfloor \tau/N \rfloor. \tag{37a}$$

Note that $u_{g,i}^{[N]}$ only depends on $\bar{e}_{\mathcal{N}_{g,i}}^{[N]} = (e_{\mathcal{N}_{g,i}}^{[N]}, e_{\mathcal{N}_{a\to g,i}}^{[N]})$ due to the control structure in Assumption 2. Hence, (17b) can be written in an equivalent overall form

$$u_{g,i}^{[N],t+1}(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l)) = \underset{u_{g,i}^{[N]}(l)}{\text{argmin}} \{L_{g,i}(l) + J_g^{t+1}(l+1)\} \tag{37b}$$

and is equivalent to the following centralized form, that is,

$$\boldsymbol{u}_{\boldsymbol{g}}^{[N],t+1}(l) = \underset{\boldsymbol{u}_{\boldsymbol{g}}^{[N]}(l)}{\text{argmin}} \{L_g(l) + J_g^{t+1}(l+1)\}. \tag{37c}$$

Note that this condition holds due to the control structure in Assumption 2.

Analogously, for the UAV formation system, one can write

$$J_a^{t+1}(\tau) = L_a(\tau) + J_a^t(\tau+1) \tag{38a}$$

$$\boldsymbol{u}_a^{t+1}(\tau) = \underset{\boldsymbol{u}_a(\tau)}{\text{argmin}} \{L_a(\tau) + J_a^{t+1}(\tau+1)\}. \tag{38b}$$

As the policy learning procedures of the UAVs and UGVs are performed in an asynchronous manner, one can utilize the proof arguments in [46] to verify that $J_g^{t+1}(\tau) \le J_g^t(\tau)$ holds in each iteration step $t+1$. With these results, one promptly has $J_a^{t+1}(\tau) \le J_a^t(\tau)$, since the learned control policies of the UGV are fixed for policy updates of UAVs. Moreover, according to [46], Claim (ii) can also be guaranteed. ■

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

TABLE II
PARAMETER SETTING TO CONDUCT NUMERICAL VALIDATION

Scenario	Platform	Δt s	$N_\star$	$\eta_{\star,i}$	$\gamma_{\star,i}$	$Q_{\star,i}$	$R_{\star,i}$
Section V-A	UAV	0.02	50	0.1	0.1	$I_{\mathcal{N}_i} \otimes \text{diag}\{12, 12, 5, 5, 2, 2\}$	$\text{diag}\{0.3, 0.3, 0.3\}$
	UGV	0.1	10	0.1	0.1	$I_{\mathcal{N}_i} \otimes \text{diag}\{1, 6, 1, 1\}$	$\text{diag}\{0.3, 0.3\}$
Section V-B	UAV	0.02	50	0.03	0.03	$I_{\mathcal{N}_i} \otimes \text{diag}\{10, 10, 10, 10, 10, 10\}$	$\text{diag}\{0.1, 0.1, 0.1\}$
	UGV	0.1	10	0.05	0.05	$I_{\mathcal{N}_i} \otimes \text{diag}\{10, 10, 10, 10\}$	$\text{diag}\{1, 1\}$
Section V-C	UAV	0.02	50	0.03	0.03	$I_{\mathcal{N}_i} \otimes \text{diag}\{10, 10, 10, 10, 10, 10\}$	$\text{diag}\{0.1, 0.1, 0.1\}$
	UGV	0.1	10	0.05	0.05	$I_{\mathcal{N}_i} \otimes \text{diag}\{10, 10, 10, 10\}$	$\text{diag}\{1, 1\}$
Section V-D	UAV	0.02	50	0.03	0.03	$I_{\mathcal{N}_i} \otimes \text{diag}\{10, 10, 10, 10, 10, 10\}$	$\text{diag}\{0.1, 0.1, 0.1\}$
	UGV	0.1	10	0.05	0.05	$I_{\mathcal{N}_i} \otimes \text{diag}\{10, 10, 10, 10\}$	$\text{diag}\{1, 1\}$
Section V-E	UAV	0.02	50	0.03	0.03	$I_{\mathcal{N}_i} \otimes \text{diag}\{10, 10, 10, 10, 10, 10\}$	$\text{diag}\{0.1, 0.1, 0.1\}$
	UGV	0.1	10	0.05	0.05	$I_{\mathcal{N}_i} \otimes \text{diag}\{10, 10, 10, 10\}$	$\text{diag}\{1, 1\}$

D. Proof to Theorem 3

Proof: The proof is organized in two steps.

1) *Stability of UAVs at the fast time scale:* Let $\overrightarrow{\boldsymbol{u}}_a^*(\tau)$ be optimal policies at fast time instant $\tau = kN$ and $\overrightarrow{\boldsymbol{u}}_a^f(\tau+1) = \boldsymbol{u}_a^*(\tau+1), \ldots, \boldsymbol{u}_a(\tau+N_a-N), \boldsymbol{u}_a(\tau+N_a-N+1), \ldots, \boldsymbol{u}_a(\tau+N_a)$ be feasible control policies at $\tau = kN+N$ such that the terminal state stays in the neighbor of the origin. Denoting $J_a^f(\tau+1)$ as the cost associated with $\overrightarrow{\boldsymbol{u}}_a^f(\tau+1)$, it follows that:

$$J_a^f(kN+N) - J_a^*(kN) = \chi_a(e_a(kN+N_a)) - \sum_{\tau=0}^{N-1} D_a(e_a(kN+\tau), \boldsymbol{u}_a^*(kN+\tau))$$

where $D_a(e_a(kN+\tau), \boldsymbol{u}_a^*(kN+\tau)) = \sum_{i=1}^{M_a}(\|\bar{e}_{\mathcal{N}_{a,i}}(kN+\tau)\|_{\bar{Q}_{a,i}}^2 + \|u_{a,i}^*(kN+\tau)\|_{R_{a,i}}^2$, and

$$\chi(e_a(kN+N)) = \sum_{\tau=0}^{N-1}\sum_{i=1}^{M_a} \|\bar{e}_{\mathcal{N}_{a,i}}(kN+\tau+N_a)\|_{\bar{Q}_{a,i}}^2 + \|u_{a,i}(kN+\tau+N_a)\|_{R_{a,i}}^2 + \|\bar{e}_{\mathcal{N}_{a,i}}(kN+N+N_a)\|_{P_{a,i}}^2 - \|\bar{e}_{\mathcal{N}_{a,i}}(kN+N_a)\|_{P_{a,i}}^2.$$

Note that, by forward evolution on the linearized model of the UAV (35), one has

$$\begin{aligned}&\|\bar{e}_{\mathcal{N}_{a,i}}(kN+N+N_a)\|_{P_{a,i}}^2 \\ &\le \|\bar{e}_{\mathcal{N}_{a,i}}(kN+N+N_a-1)\|_{F_{a,i}^\top P_{a,i} F_{a,i} + P_{a,i} S_{a,i}^2 + 2P_{a,i}F_{a,i}S_{a,i}}^2 \\ &\vdots \\ &\le \|\bar{e}_{\mathcal{N}_{a,i}}(kN+N_a)\|_{(F_{a,i}^N)^\top P_{a,i} F_{a,i}^N + \Pi(P_{a,i}, S_{a,i})}^2. \end{aligned} \tag{39}$$

Similarly, one also has

$$\begin{aligned}&\sum_{\tau=0}^{N-1} \left\|\bar{e}_{\mathcal{N}_{a,i}}(kN+\tau+N_a)\right\|_{\bar{Q}_{a,i}}^2 + \left\|u_{a,i}^*(kN+\tau+N_a)\right\|_{R_{a,i}}^2 \\ &= \sum_{\tau=0}^{N-1} \left\|\bar{e}_{\mathcal{N}_{a,i}}(kN+\tau+N_a)\right\|_{\bar{Q}_{a,i}}^2 \\ &\le \left\|\bar{e}_{\mathcal{N}_{a,i}}(kN+N_a)\right\|_{\sum_{\tau=0}^{N-1}(F_{a,i}^\tau)^\top \tilde{Q}_{a,i} F_{a,i}^\tau + \Pi(\tilde{Q}_{a,i}, S_{a,i})}^2. \end{aligned} \tag{40}$$

Consequently, in view of (32) and (36), we have $\chi_a(e_a(\tau+N)) \le 0$. In view of (32), one has

$$J_a^f(\tau+1) - J_a^*(\tau) \le -\sum_{\tau=0}^{N-1} D_a(e_a(kN+\tau), \boldsymbol{u}_a^*(kN+\tau)).$$

Hence, e_a and $\boldsymbol{u}_a$ converge to the origin asymptotically.

2) Stability of UGVs at the slow time scale: Let $\overrightarrow{\boldsymbol{u}}_g^*(l)$ be optimal policies at slow time instant l and $\overrightarrow{\boldsymbol{u}}_g^f(l+1) = \boldsymbol{u}_g^*(l+1), \ldots, \boldsymbol{u}_g^*(l+N_g-1), \boldsymbol{u}_g(l+N_g)$ be such that the terminal state stays in the neighbor of the origin. Denoting $J_g^f(l+1)$ as the cost associated with $\overrightarrow{\boldsymbol{u}}_g^f(l+1)$, it follows that:

$$\begin{aligned}&J_g^f(l+1) - J_g^*(l) \\ &= -D_g(e_g^{[N]}(l), \boldsymbol{u}_g^{[N],*}(l)) + \chi_g(e_g^{[N]}(l+N_g))\end{aligned}$$

where $D_g(e_g^{[N]}(l), \boldsymbol{u}_g{}^{[N],*}(l)) = \sum_{i=1}^{M_g}(\|\bar{e}_{\mathcal{N}_{g,i}}(l)\|_{\bar{Q}_{g,i}}^2 + \|u_i^{[N],*}(l)\|_{R_{g,i}}^2$, $\chi(e_g^{[N]}(l+N_g)) = \sum_{i=1}^{M_g} \|\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l+N_g)\|_{\bar{Q}_{g,i}}^2 + \|u_{g,i}^{[N],*}(l+N_g)\|_{R_{g,i}}^2 + \|\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l+N_g+1)\|_{P_{g,i}}^2 - \|\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l+N_g)\|_{P_{g,i}}^2$. Note that, by evolution on the linearized model (33), one has

$$\begin{aligned}&\|\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l+N_g+1)\|_{P_{g,i}}^2 = \|\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l+N_g)\|_{F_{g,i}^\top P_{g,i} F_{g,i}}^2 \\ &+ \|\text{Er}_{g,i}(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l+N_g)\|_{P_{g,i}}^2 \\ &+ 2\text{Er}_{g,i}(\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l+N_g))^\top P_{g,i} \bar{e}_{\mathcal{N}_{g,i}}(l+N_g) \\ &\le \|\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l+N_g)\|_{F_{g,i}^\top P_{g,i} F_{g,i}}^2 + (\|P_{g,i}\| S_{g,i}^2 \\ &+ 2\|P_{g,i}F_{g,i}\|S_{g,i})\|\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l+N_g)\|^2 \\ &\le \|\bar{e}_{\mathcal{N}_{g,i}}^{[N]}(l+N_g)\|_{F_{g,i}^\top P_{g,i} F_{g,i} + (\beta_{g,i}-1)\bar{Q}_{g,i}}^2 \end{aligned} \tag{41}$$

where the last inequality is derived with (36). Consequently, in view of (32), we have $\chi_g(e_g^{[N]}(l+N)) \le 0$. In view of (32), one has

$$J_g^f(l+1) - J_g^*(l) \le -D_g(e_g^{[N]}(l), \boldsymbol{u}_g^{[N]}(l))$$

Hence, $e_g^{[N]}$ and $\boldsymbol{u}_g^{[N]}$ converge to the origin asymptotically. ■

E. Simulation Settings

The parameter settings to conduct the numerical validation presented in Section V are presented here.

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

1) Coordinate Correction Variables Used in Section V-A:

$$s_{\star,r} = \left[10\cos(0.1t-\pi/2), 11+10\sin(0.1t-\pi/2)\right]^\top,$$
$$\star = a, g$$
$$\rho_{\star,ri} = \left[\cos\varphi_{\star,i} d^x_{\star,ri} - \sin\varphi_{\star,i} d^y_{\star,ri} \quad \sin\varphi_{\star,i} d^x_{\star,ri} + \cos\varphi_{\star,i} d^y_{\star,ri}\right]^\top$$
$$\rho_{\star,ji} = \left[\cos\varphi_i d^x_{\star,ji} - \sin\varphi_i d^y_{\star,ji} \quad \sin\varphi_i d^x_{\star,ji} + \cos\varphi_i d^y_{\star,ji}\right]^\top$$
$$d_{g,r1} = \left[d^x_{g,r1} \; d^y_{g,r1}\right]^\top = [0 \quad 0]^\top$$
$$d_{a,r1} = \left[d^x_{a,r1} \; d^y_{a,r1}\right]^\top = [0 \quad -20]^\top$$
$$d_{g,ji} = \left[d^x_{g,ji} \; d^y_{g,ji}\right]^\top = [0 \; -2 \; -2 \; 0 \; 2 \; 2; -2 \; 0 \; 0 \; 2 \; 0 \; 0]$$
$$d_{a,ji} = \left[d^x_{a,ji} \; d^y_{a,ji}\right]^\top = [0 \quad -2]^\top$$

2) Coordinate Correction Variables Used in Section V-B:

$$s_{g,r} = \left[v_r t + 15, 7.5\right]^\top, v_r = 1$$
$$s_{a,r} = \left[v_r t + 30\cos(0.1t-\pi/2), 30\sin(0.1t-\pi/2)\right]^T$$
$$\rho_{\star,r1} = [0 \; 0]^\top, \star = a, g$$
$$\rho_{g,ji} = [0 \; -15 \; -15 \; 0 \; 15 \; 15; -15 \; 0 \; 0 \; 15 \; 0 \; 0]$$
$$\rho_{a,ji} = [0 \; 5]^\top.$$

3) Disturbance Used in Section V-D: UAV disturbance: $0.2\exp^{-5t}\sin(10t)$ in x, y, z axes; and UGV disturbance: $[0.2\exp^{-t}\cos(5t), 0.1\exp^{-t}\sin(5t)]$ in x, y axes.

4) Algorithmic Parameters Used in Section V: The detailed algorithmic parameter settings to conduct the numerical validations are presented in Table II.

REFERENCES

[1] H. Tang, Y. Chen, and I. Ali, "Cross-dimensional distributed control for heterogeneous UAV-UGV systems with nonzero leader input," *IEEE Trans. Intell. Veh.*, vol. 9, no. 10, pp. 6354–6368, Oct. 2024.

[2] P. Guo, R. Zhang, and B. Xu, "Safety separation distance design for UAV formation based on system performance," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 3, pp. 5980–5995, Jun. 2025.

[3] O. S. Oubbati, J. Alotaibi, F. Alromithy, M. Atiquzzaman, and M. R. Altimania, "A UAV-UGV cooperative system: Patrolling and energy management for urban monitoring," *IEEE Trans. Veh. Technol.*, early access, Apr. 24, 2025, doi: 10.1109/TVT.2025.3563971.

[4] N. Thelasingha, A. A. Julius, J. Humann, J.-P. Reddinger, J. Dotterweich, and M. Childers, "Iterative planning for multi-agent systems: An application in energy-aware UAV-UGV cooperative task site assignments," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 3685–3703, 2025.

[5] X. Guo, Q. Li, Q. Yao, and X. Lu, "Fixed-time dynamic event-triggered formation control for unmanned aerial-ground vehicle (UAV-UGV) heterogeneous system," *Int. J. Robust Nonlinear Control*, vol. 24, pp. 79–87, 2025.

[6] M. Farina, X. Zhang, and R. Scattolini, "A hierarchical multi-rate MPC scheme for interconnected systems," *Automatica*, vol. 90, pp. 38–46, 2018.

[7] J. Li, T. Sun, X. Huang, L. Ma, and V. C. M. Leung, "A memetic path planning algorithm for unmanned air/ground vehicle cooperative detection systems," *IEEE Trans. Automat. Sci. Eng.*, vol. 19, no. 4, pp. 2724–2737, Apr. 2022.

[8] T. W. Nguyen, L. Catoire, and E. Garone, "Control of a quadrotor and a ground vehicle manipulating an object," *Automatica*, vol. 105, pp. 384–390, 2019.

[9] Y. Hua, X. Dong, G. Hu, Q. Li, and Z. Ren, "Distributed time-varying output formation tracking for heterogeneous linear multiagent systems with a nonautonomous leader of unknown input," *IEEE Trans. Autom. Control*, vol. 64, no. 10, pp. 4292–4299, Oct. 2019.

[10] S. Zhao, Z. Li, and Z. Ding, "Bearing-only formation tracking control of multiagent systems," *IEEE Trans. Autom. Control*, vol. 64, no. 11, pp. 4541–4554, Nov. 2019.

[11] P. Chen, L. Luo, D. Guo, X. Luo, X. Li, and Y. Sun, "Secure task offloading for rural area surveillance based on UAV-UGV collaborations," *IEEE Trans. Veh. Technol.*, vol. 73, no. 1, pp. 923–937, Jan. 2023.

[12] I. Ravanshadi, E. A. Boroujeni, and M. Pourgholi, "Centralized and distributed model predictive control for consensus of non-linear multi-agent systems with time-varying obstacle avoidance," *ISA Trans.*, vol. 133, pp. 75–90, 2023.

[13] Z. Zhou and H. Xu, "Decentralized adaptive optimal tracking control for massive autonomous vehicle systems with heterogeneous dynamics: A stackelberg game," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 12, pp. 5654–5663, Dec. 2021.

[14] H. Liang, S. Yang, T. Li, and H. Zhang, "Distributed adaptive cooperative control for human-in-the-loop heterogeneous UAV-UGV systems with prescribed performance," *IEEE Trans. Intell. Veh.*, vol. 9, no. 11, pp. 6912–6925, Nov. 2024.

[15] Y. Ma, B. Jiang, J. Wang, and J. Gong, "Adaptive fault-tolerant formation control for heterogeneous UAVS-UGVS systems with multiple actuator faults," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 5, pp. 6705–6716, Oct. 2023.

[16] C. Chen, K. Xie, F. L. Lewis, S. Xie, and A. Davoudi, "Fully distributed resilience for adaptive exponential synchronization of heterogeneous multiagent systems against actuator faults," *IEEE Trans. Autom. Control*, vol. 64, no. 8, pp. 3347–3354, Aug. 2018.

[17] J. Ding, F. Marcassa, S.-C. Wu, and M. Tomizuka, "Multirate control for computation saving," *IEEE Trans. Control Syst. Technol.*, vol. 14, no. 1, pp. 165–169, Jan. 2005.

[18] J. Li, T. Chai, F. L. Lewis, J. Fan, Z. Ding, and J. Ding, "Off-policy q-learning: Set-point design for optimizing dual-rate rougher flotation operational processes," *IEEE Trans. Ind. Electron.*, vol. 65, no. 5, pp. 4092–4102, May 2017.

[19] Z. Xia, M. Hu, W. Dai, H. Yan, and X. Ma, "Q-learning-based multi-rate optimal control for process industries," *IEEE Trans. Circuits Syst. II: Exp. Briefs*, vol. 70, no. 6, pp. 2006–2010, Jun. 2022.

[20] W. Dai, T. Li, L. Zhang, Y. Jia, and H. Yan, "Multi-rate layered operational optimal control for large-scale industrial processes," *IEEE Trans. Ind. Informat.*, vol. 18, no. 7, pp. 4749–4761, Jul. 2021.

[21] J. Li, M. Yang, and F. L. Lewis, "Optimal operational self-learning control for multi-time scale industrial processes with signal compensations," *Eng. Appl. Artif. Intell.*, vol. 126, 2023, Art. no. 107065.

[22] A. Sala, A. Cuenca, and J. Salt, "A retunable PID multi-rate controller for a networked control system," *Inf. Sciences*, vol. 179, no. 14, pp. 2390–2402, 2009.

[23] F. Ferrante, R. G. Sanfelice, and S. Tarbouriech, "Control design under actuator saturation and multi-rate sampling," *Automatica*, vol. 148, 2023, Art. no. 110767.

[24] M. De Stefano, H. Mishra, R. Balachandran, R. Lampariello, C. Ott, and C. Secchi, "Multi-rate tracking control for a space robot on a controlled satellite: A passivity-based strategy," *IEEE Robot. Automat. Lett.*, vol. 4, no. 2, pp. 1319–1326, Feb. 2019.

[25] U. Rosolia and A. D. Ames, "Multi-rate control design leveraging control barrier functions and model predictive control policies," *IEEE Contr. Syst. Lett.*, vol. 5, no. 3, pp. 1007–1012, Mar. 2020.

[26] E. Terzi, M. Farina, L. Fagiano, and R. Scattolini, "Robust multi-rate predictive control using multi-step prediction models learned from data," *Automatica*, vol. 136, 2022, Art. no. 109852.

[27] K. Garg, R. K. Cosner, U. Rosolia, A. D. Ames, and D. Panagou, "Multi-rate control design under input constraints via fixed-time barrier functions," *IEEE Contr. Syst. Lett.*, vol. 6, pp. 608–613, 2021.

[28] H. Han, S. Fu, H. Sun, and J. Qiao, "Data-driven multimodel predictive control for multirate sampled-data nonlinear systems," *IEEE Trans. Autom. Sci. Eng.*, vol. 20, no. 3, pp. 2182–2194, Jul. 2022.

[29] J. Yang, C. Liu, M. Coombes, Y. Yan, and W.-H. Chen, "Optimal path following for small fixed-wing UAVS under wind disturbances," *IEEE Trans. Control Syst. Technol.*, vol. 29, no. 3, pp. 996–1008, Mar. 2020.

[30] S.-X. Xiong, X.-P. Xie, and G.-P. Jiang, "Fully distributed optimal tracking control for UAV-UGV formation against byzantine attacks," *IEEE Trans. Veh. Technol.*, early access, Apr. 30, 2025, doi: 10.1109/TVT.2025.3565765.

[31] M. Schwenzer, M. Ay, T. Bergs, and D. Abel, "Review on model predictive control: An engineering perspective," *Int. J. Adv. Manuf. Technol.*, vol. 117, no. 5, pp. 1327–1349, 2021.

[32] H. Li and Y. Shi, "Robust distributed model predictive control of constrained continuous-time nonlinear systems: A robustness constraint approach," *IEEE Trans. Autom. Control*, vol. 59, no. 6, pp. 1673–1678, Jun. 2013.

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.

[33] R. R. Negenborn and J. M. Maestre, "Distributed model predictive control: An overview and roadmap of future research opportunities," *EEE Control Syst. Mag.*, vol. 34, no. 4, pp. 87–97, Aug. 2014.
[34] C. Xu, S. Liu, C. Zhang, Y. Huang, Z. Lu, and L. Yang, "Multi-agent reinforcement learning based distributed transmission in collaborative cloud-edge systems," *IEEE Trans. Veh. Technol.*, vol. 70, no. 2, pp. 1658–1672, Feb. 2021.
[35] M. S. Allahham, A. A. Abdellatif, N. Mhaisen, A. Mohamed, A. Erbad, and M. Guizani, "Multi-agent reinforcement learning for network selection and resource allocation in heterogeneous multi-RAT networks," *IEEE Trans. Cogn. Commun. Netw.*, vol. 8, no. 2, pp. 1287–1300, Feb. 2022.
[36] B. Yan, P. Shi, C.-C. Lim, and Z. Shi, "Optimal robust formation control for heterogeneous multi-agent systems based on reinforcement learning," *Int. J. Robust Nonlinear Control*, vol. 32, no. 5, pp. 2683–2704, May 2022.
[37] X. Zhang, J. Liu, X. Xu, S. Yu, and H. Chen, "Robust learning-based predictive control for discrete-time nonlinear systems with unknown dynamics and state constraints," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 52, no. 12, pp. 7314–7327, Dec. 2022.
[38] X. Zhang, Y. Jiang, Y. Lu, and X. Xu, "Receding-horizon reinforcement learning approach for kinodynamic motion planning of autonomous vehicles," *IEEE Trans. Intell. Veh.*, vol. 7, no. 3, pp. 556–568, Mar. 2022.
[39] X. Zhang et al., "Toward scalable multirobot control: Fast policy learning in distributed MPC," *IEEE Trans. Robot.*, vol. 41, pp. 1491–1512, 2025.
[40] C. Conte, C. N. Jones, M. Morari, and M. N. Zeilinger, "Distributed synthesis and stability of cooperative distributed model predictive control for linear systems," *Automatica*, vol. 69, pp. 117–125, 2016.
[41] S. Liang, F. Wang, Z. Chen, and Z. Liu, "Formation control for discrete-time heterogeneous multi-agent systems," *Int. J. Robust Nonlinear Control*, vol. 32, no. 10, pp. 5848–5865, Oct. 2022.
[42] W. Zhao, H. Liu, Y. Wan, and Z. Lin, "Data-driven formation control for multiple heterogeneous vehicles in air-ground coordination," *IEEE Control Netw. Syst.*, vol. 9, no. 4, pp. 1851–1862, Apr. 2022.
[43] K. Xiao, L. Ma, S. Tan, Y. Cong, and X. Wang, "Implementation of UAV coordination based on a hierarchical multi-UAV simulation platform," in *Advances in Guid., Navigation Control: Proc. 2020 Int. Conf. Guid., Navigation Control (ICGNC), Tianjin, China, October 23–25, 2020.* New York: Springer, 2022, pp. 5131–5143.
[44] P. Mhaskar, N. H. El-Farra, and P. D. Christofides, "Stabilization of nonlinear systems with state and control constraints using lyapunov-based predictive control," *Syst. Control Lett.*, vol. 55, no. 8, pp. 650–659, 2006.
[45] D. Chmielewski and V. Manousiouthakis, "On constrained infinite-time linear quadratic optimal control," *Syst. Control Lett.*, vol. 29, no. 3, pp. 121–129, 1996.
[46] X. Zhang, Y. Peng, B. Luo, W. Pan, X. Xu, and H. Xie, "Model-based safe reinforcement learning with time-varying constraints: Applications to intelligent vehicles," *IEEE Trans. Ind. Electron.*, vol. 71, no. 10, pp. 12744–12753, Oct. 2024.



Xinglong Zhang (Member, IEEE) received the B.E. degree in mechanical engineering from Zhejiang University, Hangzhou, China, in 2011, and the Ph.D. degree in system and control from Politecnico di Milano, Milan, Italy, in 2018.

He is currently an Associate Professor with the College of Intelligence Science and Technology, National University of Defense Technology, Changsha, China. His research interests include Koopman operators, learning-based model predictive control, reinforcement learning and approximate dynamic programming, and their applications in automotive systems.

Dr. Zhang is on the Editorial Board for *Discover Vehicles* in Springer Nature.



Cong Li received the Ph.D. degree in learning-based control and robotics from the Chair of Automatic Control Engineering, Technical University of Munich, Munich, Germany, in 2022.

He is currently a Postdoctoral Fellow with the National University of Defense Technology, Changsha, China. His research interests include reinforcement learning, optimal control, robust control, constraint optimization, and robotics.



Ronghua Zhang received the M.S. degree in instruments science and technology from Chongqing University, Chongqing, China, in 2013. She is currently working toward the Ph.D. degree in control science and engineering with the National University of Defense Technology, Changsha, China.

Since 2013, she has been Lecturer with Sichuan University of Science & Engineering, Sichuan, China. Her research interests include formation control, reinforcement learning, and multirobot systems.



Quan Xiong received the Ph.D. degree in control science and engineering from the School of Automation, Central South University, Changsha, China, in 2020.

He is currently an Assistant Professor with the National University of Defense Technology, Changsha, China. His research interests include the distributed control robust control and learning-based control.



Wei Jiang received the Ph.D. degree in instrument science and technology from the National University of Defense Technology (NUDT), Changsha, China, in 2017.

Since 2023, she has been an Associate Professor with the College of Intelligence Science and Technology, NUDT. Her research interests include the theory and technology of smart measurement and metrology, Koopman operators, and unmanned systems.



Xin Xu (Senior Member, IEEE) received the Ph.D. degree in control science and engineering from the College of Mechatronics and Automation, National University of Defense Technology (NUDT), Changsha, China, in 2002.

He has been a Visiting Professor with the Hong Kong Polytechnic University, Hong Kong, China. He is currently a Professor with the College of Intelligence Science and Technology, NUDT. His research interests include intelligent control, reinforcement learning, robotics, and autonomous vehicles.

He has served as an Associate Editor and a Guest Editor for *Information Sciences*, IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS: SYSTEMS, INTERNATIONAL JOURNAL OF ADAPTIVE CONTROL AND SIGNAL PROCESSING, and *Acta Automatica Sinica*.

Authorized licensed use limited to: UNIVERSITE DE GAFSA. Downloaded on June 11,2026 at 22:49:44 UTC from IEEE Xplore. Restrictions apply.